

DSS150P Lab Activity #3

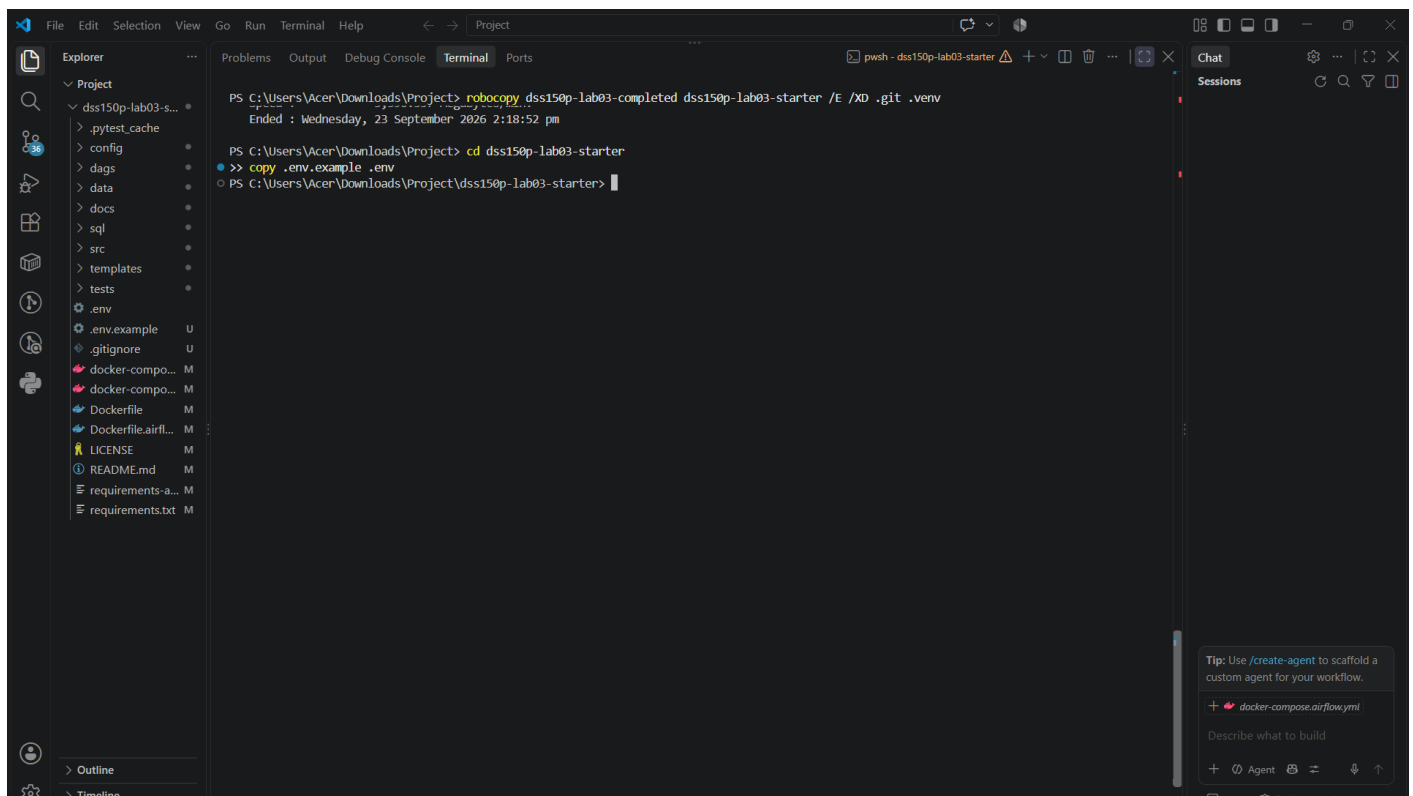
Productionizing a Modular Data Pipeline

Run Evidence - Screenshots

Chronological evidence of environment setup, pipeline execution, PostgreSQL loading, testing, benchmarking, and Airflow orchestration.

Screenshots cropped to remove browser tabs / personal information.

Checkpoint 1 - Cloning starter repo



Checkpoint 1 - Copying completed files into clone (robocopy)

File Edit Selection View Go Run Terminal Help

Explorer

- Project
 - dss150p-lab03-start
 - .pytest_cache
 - .venv
 - config
 - dags
 - data
 - docs
 - sql
 - src
 - templates
 - tests
 - .env
 - .env.example
 - .gitignore
 - docker-compose.yml
 - docker-compose-airflow.yml
 - Dockerfile
 - LICENSE
 - README.md
 - requirements-airflow.txt
 - requirements.txt

Problems Output Debug Console Terminal Ports

Terminal

```
PS C:\Users\Acer\Downloads\Project> robocopy dss150p-lab03-completed dss150p-lab03-start /E /XD .git .venv
Speed : 3,390.357 Megabytes/min.
Ended : Wednesday, 23 September 2026 2:18:52 pm

PS C:\Users\Acer\Downloads\Project> cd dss150p-lab03-start
>> copy .env.example .env
PS C:\Users\Acer\Downloads\Project\dss150p-lab03-start> python -m venv .venv
>> .venv\Scripts\Activate.ps1
>> pip install -r requirements.txt
Traceback (most recent call last):
  File "<frozen runpy>", line 203, in _run_module_as_main
  File "<frozen runpy>", line 88, in _run_code
  File "C:\Users\Acer\AppData\Local\Python\pythoncore-3.14-64\Lib\venv\__main__.py", line 6, in <module>
    main()
    ~~~~
  File "C:\Users\Acer\AppData\Local\Python\pythoncore-3.14-64\Lib\venv\__init__.py", line 690, in main
    builder.create(d)
    ~~~~~
  File "C:\Users\Acer\AppData\Local\Python\pythoncore-3.14-64\Lib\venv\__init__.py", line 82, in create
    self._setup_pip(context)
    ~~~~~
  File "C:\Users\Acer\AppData\Local\Python\pythoncore-3.14-64\Lib\venv\__init__.py", line 458, in _setup_pip
    self._call_new_python(context, '-m', 'ensurepip', '--upgrade',
    ~~~~~
    '--default-pip', stderr=subprocess.STDOUT)
    ~~~~~
  File "C:\Users\Acer\AppData\Local\Python\pythoncore-3.14-64\Lib\venv\__init__.py", line 454, in _call_new_python
    subprocess.check_output(args, **kwargs)
    ~~~~~
  File "C:\Users\Acer\AppData\Local\Python\pythoncore-3.14-64\Lib\subprocess.py", line 473, in check_output
    return run(*popenargs, stdout=PIPE, timeout=timeout, check=True,
    ~~~~~
    **kwargs).stdout
    ~~~~~
  File "C:\Users\Acer\AppData\Local\Python\pythoncore-3.14-64\Lib\subprocess.py", line 557, in run
    stdout, stderr = process.communicate(input, timeout=timeout)
    ~~~~~
  File "C:\Users\Acer\AppData\Local\Python\pythoncore-3.14-64\Lib\subprocess.py", line 1288, in communicate
    stdout = self.stdout.read()
KeyboardInterrupt
PS C:\Users\Acer\Downloads\Project\dss150p-lab03-start>
```

Chat

Sessions

Tip: Use `/create-agent` to scaffold a custom agent for your workflow.

+ `docker-compose.airflow.yml`

Describe what to build

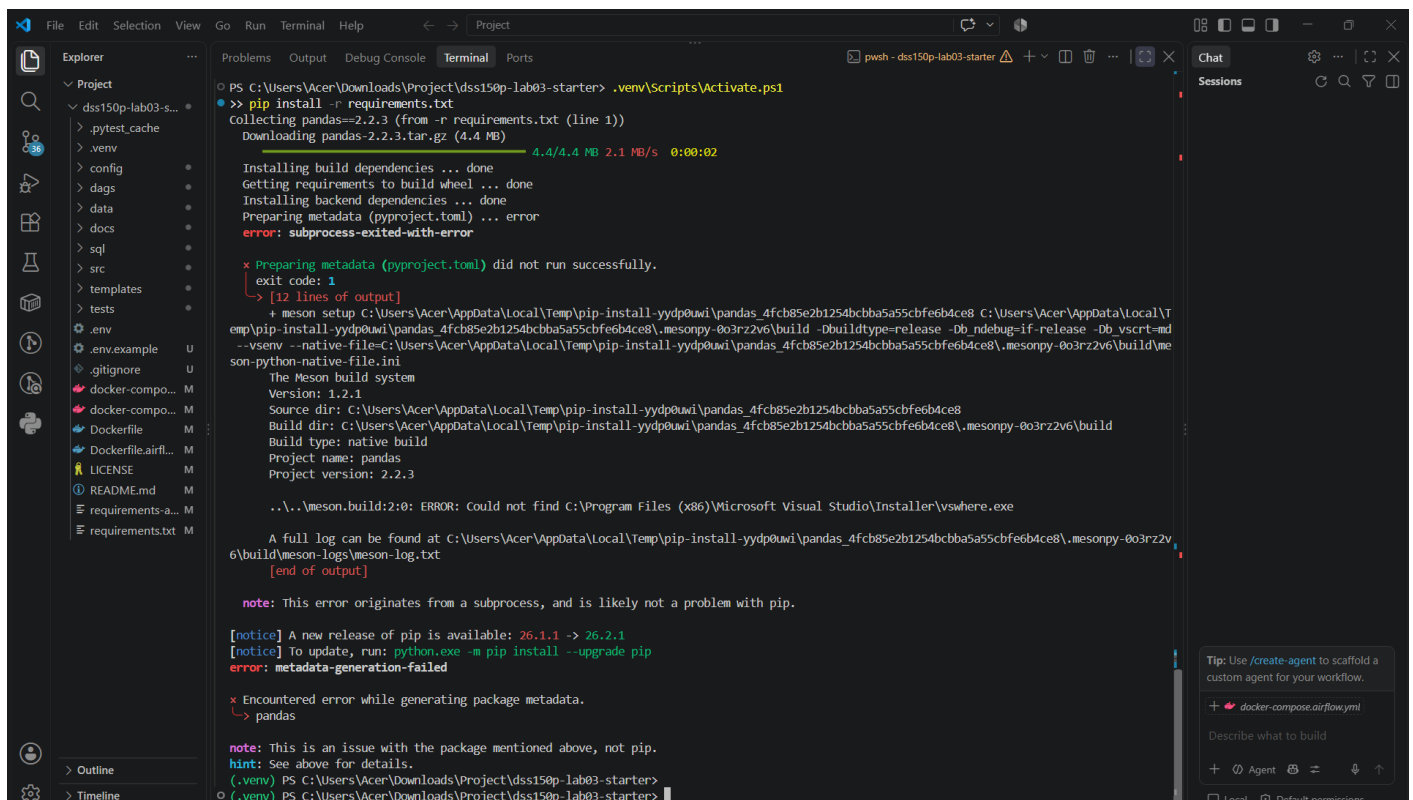
+

Local Default permissions

Local main* Launchpad 0 0 0

Jay Ryan Mapanao, 5 days ago Ln 1, Col 1 Spaces: 2 UTF-8 LF [] Compose []

Checkpoint 1 - venv creation interrupted (retry needed)



```
PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> .venv\Scripts\Activate.ps1
>> pip install -r requirements.txt
collecting pandas==2.2.3 (from -r requirements.txt (line 1))
  Downloading pandas-2.2.3.tar.gz (4.4 MB)
    4.4/4.4 MB 2.1 MB/s 0:00:02
Installing build dependencies ... done
Getting requirements to build wheel ... done
Installing backend dependencies ... done
Preparing metadata (pyproject.toml) ... error
error: subprocess-exited-with-error

x Preparing metadata (pyproject.toml) did not run successfully.
  exit code: 1
  > [12 lines of output]
+ meson setup C:\Users\Acer\AppData\Local\Temp\pip-install-yyp0uwi\pandas_4fcb85e2b1254bcbb5a55cbfe6b4ce8 C:\Users\Acer\AppData\Local\Temp\pip-install-yyp0uwi\pandas_4fcb85e2b1254bcbb5a55cbfe6b4ce8\mesonpy-003r2zv6\build -Dbuildtype=release -Db_ndebug-if-release -Db_yscrt-md
--vsenv --native-file=C:\Users\Acer\AppData\Local\Temp\pip-install-yyp0uwi\pandas_4fcb85e2b1254bcbb5a55cbfe6b4ce8\mesonpy-003r2zv6\build\meson-python-native-file.ini
The Meson build system
Version: 1.2.1
Source dir: C:\Users\Acer\AppData\Local\Temp\pip-install-yyp0uwi\pandas_4fcb85e2b1254bcbb5a55cbfe6b4ce8
Build dir: C:\Users\Acer\AppData\Local\Temp\pip-install-yyp0uwi\pandas_4fcb85e2b1254bcbb5a55cbfe6b4ce8\mesonpy-003r2zv6\build
Build type: native build
Project name: pandas
Project version: 2.2.3

..\.meson.build:2:0: ERROR: Could not find C:\Program Files (x86)\Microsoft Visual Studio\Installer\vswhere.exe

A full log can be found at C:\Users\Acer\AppData\Local\Temp\pip-install-yyp0uwi\pandas_4fcb85e2b1254bcbb5a55cbfe6b4ce8\mesonpy-003r2zv6\build\meson-logs\meson-log.txt
[end of output]

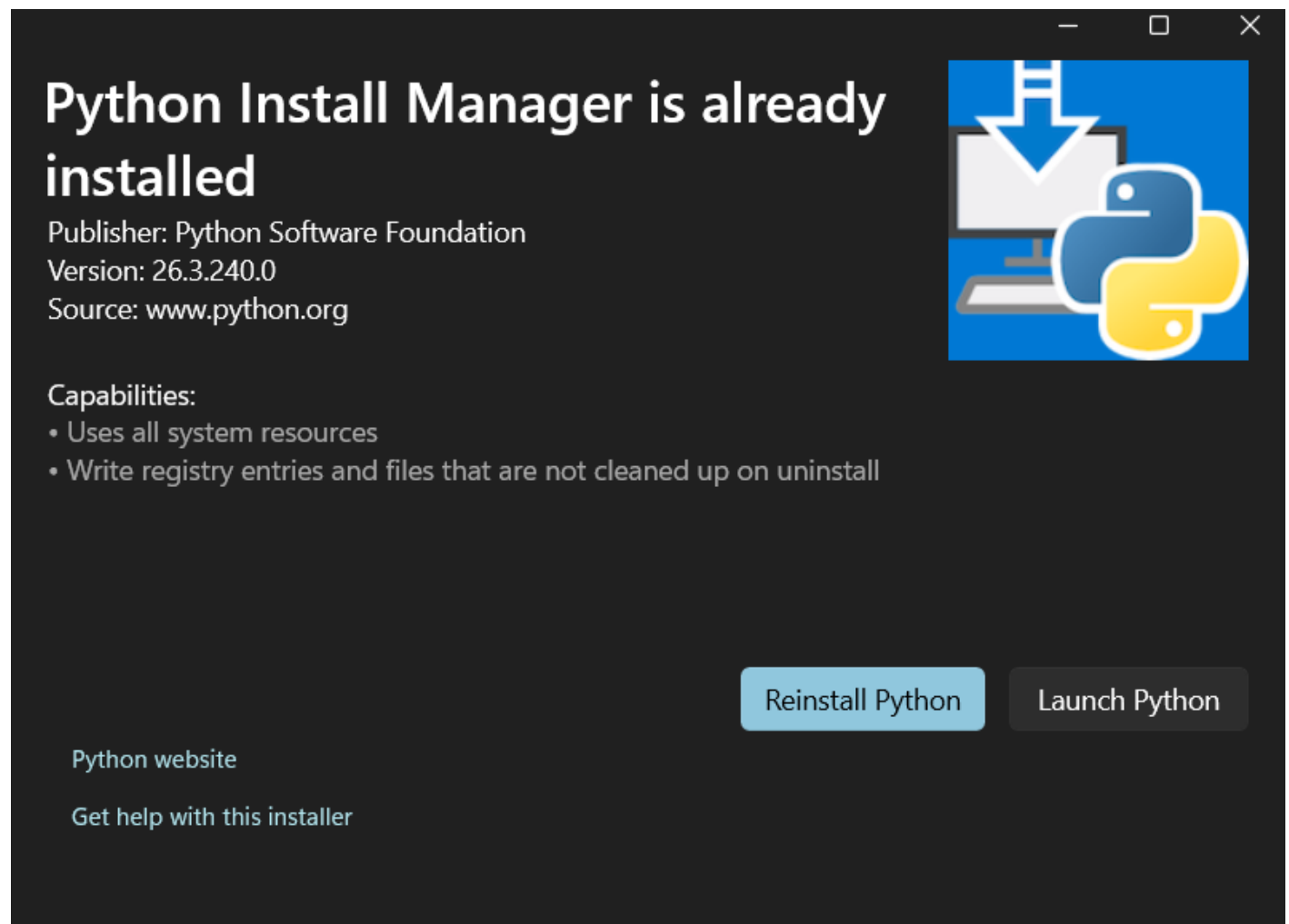
note: This error originates from a subprocess, and is likely not a problem with pip.

[notice] A new release of pip is available: 26.1.1 -> 26.2.1
[notice] to update, run: python.exe -m pip install --upgrade pip
error: metadata-generation-failed

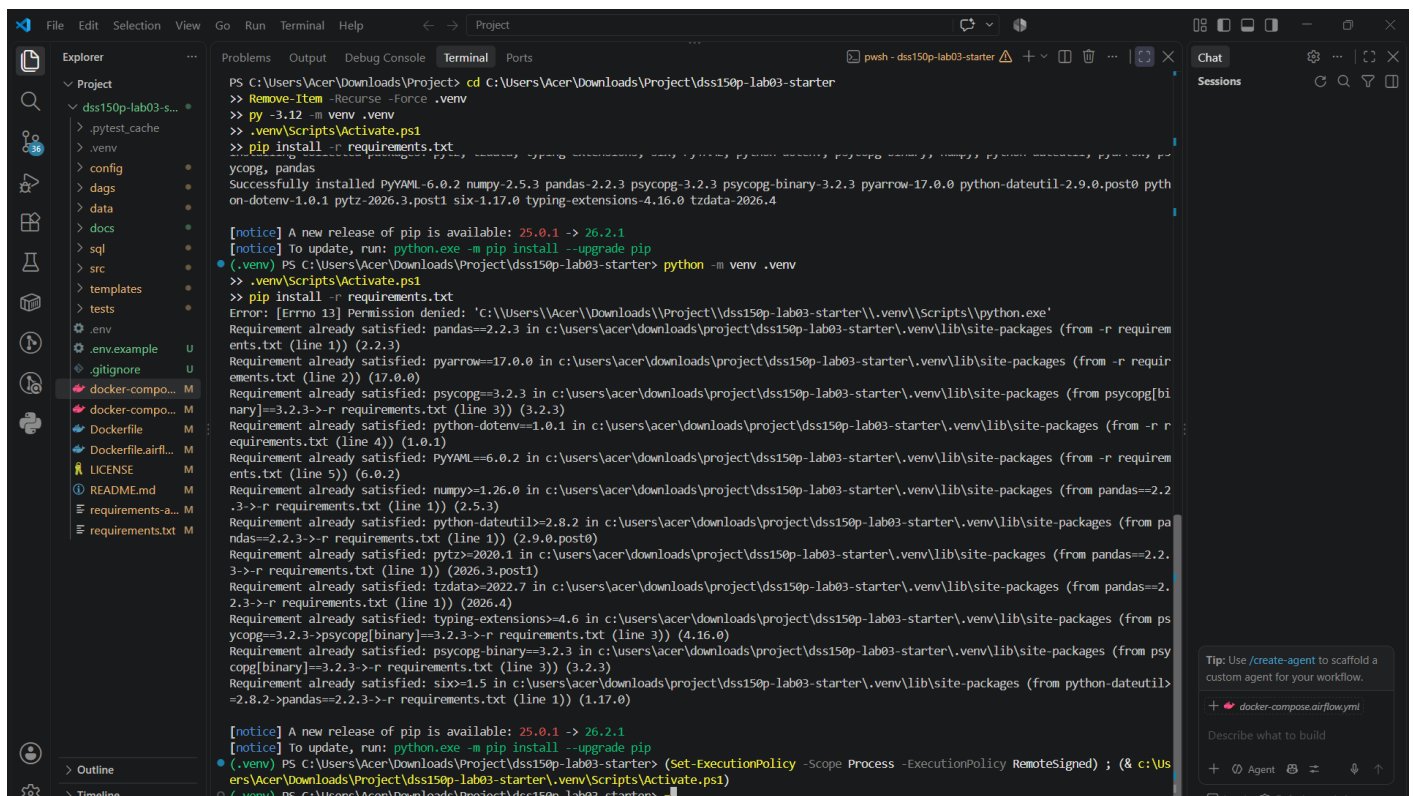
x Encountered error while generating package metadata.
  pandas

note: This is an issue with the package mentioned above, not pip.
hint: See above for details.
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter>
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter>
```

Checkpoint 1 - Installing Python 3.12 (compatibility fix)



Checkpoint 1 - Dependencies installed successfully on Python 3.12



```
PS C:\Users\Acer\Downloads\Project> cd C:\Users\Acer\Downloads\Project\dss150p-lab03-starter
>> Remove-Item -Recurse -Force .venv
>> py -3.12 -m venv .venv
>> .venv\Scripts\Activate.ps1
>> pip install -r requirements.txt
ycopp, pandas
Successfully installed PyYAML-6.0.2 numpy-2.5.3 pandas-2.2.3 psycopp-3.2.3 psycopp-binary-3.2.3 pyarrow-17.0.0 python-dateutil-2.9.0.post0 pyth
on-dotenv-1.0.1 pytz-2026.3.post1 six-1.17.0 typing-extensions-4.16.0 tzdata-2026.4

[notice] A new release of pip is available: 25.0.1 -> 26.2.1
[notice] To update, run: python.exe -m pip install --upgrade pip
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> python -m venv .venv
>> .venv\Scripts\Activate.ps1
>> pip install -r requirements.txt
Error: [Errno 13] Permission denied: 'c:\Users\Acer\Downloads\Project\dss150p-lab03-starter\.venv\Scripts\python.exe'
Requirement already satisfied: pandas==2.2.3 in c:\users\acer\downloads\project\dss150p-lab03-starter\.venv\lib\site-packages (from -r requirem
ents.txt (line 1)) (2.2.3)
Requirement already satisfied: pyarrow==17.0.0 in c:\users\acer\downloads\project\dss150p-lab03-starter\.venv\lib\site-packages (from -r requir
ements.txt (line 2)) (17.0.0)
Requirement already satisfied: psycopp==3.2.3 in c:\users\acer\downloads\project\dss150p-lab03-starter\.venv\lib\site-packages (from psycopp[bi
nary]==3.2.3->-r requirements.txt (line 3)) (3.2.3)
Requirement already satisfied: python-dotenv==1.0.1 in c:\users\acer\downloads\project\dss150p-lab03-starter\.venv\lib\site-packages (from -r r
equirements.txt (line 4)) (1.0.1)
Requirement already satisfied: PyYAML==6.0.2 in c:\users\acer\downloads\project\dss150p-lab03-starter\.venv\lib\site-packages (from -r requirem
ents.txt (line 5)) (6.0.2)
Requirement already satisfied: numpy>=1.26.0 in c:\users\acer\downloads\project\dss150p-lab03-starter\.venv\lib\site-packages (from pandas==2.2
.3->-r requirements.txt (line 1)) (2.5.3)
Requirement already satisfied: python-dateutil>=2.8.2 in c:\users\acer\downloads\project\dss150p-lab03-starter\.venv\lib\site-packages (from pa
ndas==2.2.3->-r requirements.txt (line 1)) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in c:\users\acer\downloads\project\dss150p-lab03-starter\.venv\lib\site-packages (from pandas==2.2
.3->-r requirements.txt (line 1)) (2026.3.post1)
Requirement already satisfied: tzdata>=2022.7 in c:\users\acer\downloads\project\dss150p-lab03-starter\.venv\lib\site-packages (from pandas==2.
2.3->-r requirements.txt (line 1)) (2026.4)
Requirement already satisfied: typing-extensions>=4.6 in c:\users\acer\downloads\project\dss150p-lab03-starter\.venv\lib\site-packages (from ps
ycopp==3.2.3->psycopp[binary]==3.2.3->-r requirements.txt (line 3)) (4.16.0)
Requirement already satisfied: psycopp-binary==3.2.3 in c:\users\acer\downloads\project\dss150p-lab03-starter\.venv\lib\site-packages (from psy
copp[binary]==3.2.3->-r requirements.txt (line 3)) (3.2.3)
Requirement already satisfied: six>=1.5 in c:\users\acer\downloads\project\dss150p-lab03-starter\.venv\lib\site-packages (from python-dateutil>
=2.8.2->pandas==2.2.3->-r requirements.txt (line 1)) (1.17.0)

[notice] A new release of pip is available: 25.0.1 -> 26.2.1
[notice] To update, run: python.exe -m pip install --upgrade pip
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> (Set-ExecutionPolicy -Scope Process -ExecutionPolicy RemoteSigned) ; (& c:\Us
ers\Acer\Downloads\Project\dss150p-lab03-starter\.venv\Scripts\Activate.ps1)
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter>
```

Checkpoint 1 - Python version and import check confirmed

The screenshot shows a VS Code interface with a terminal window open. The terminal displays the following commands and output:

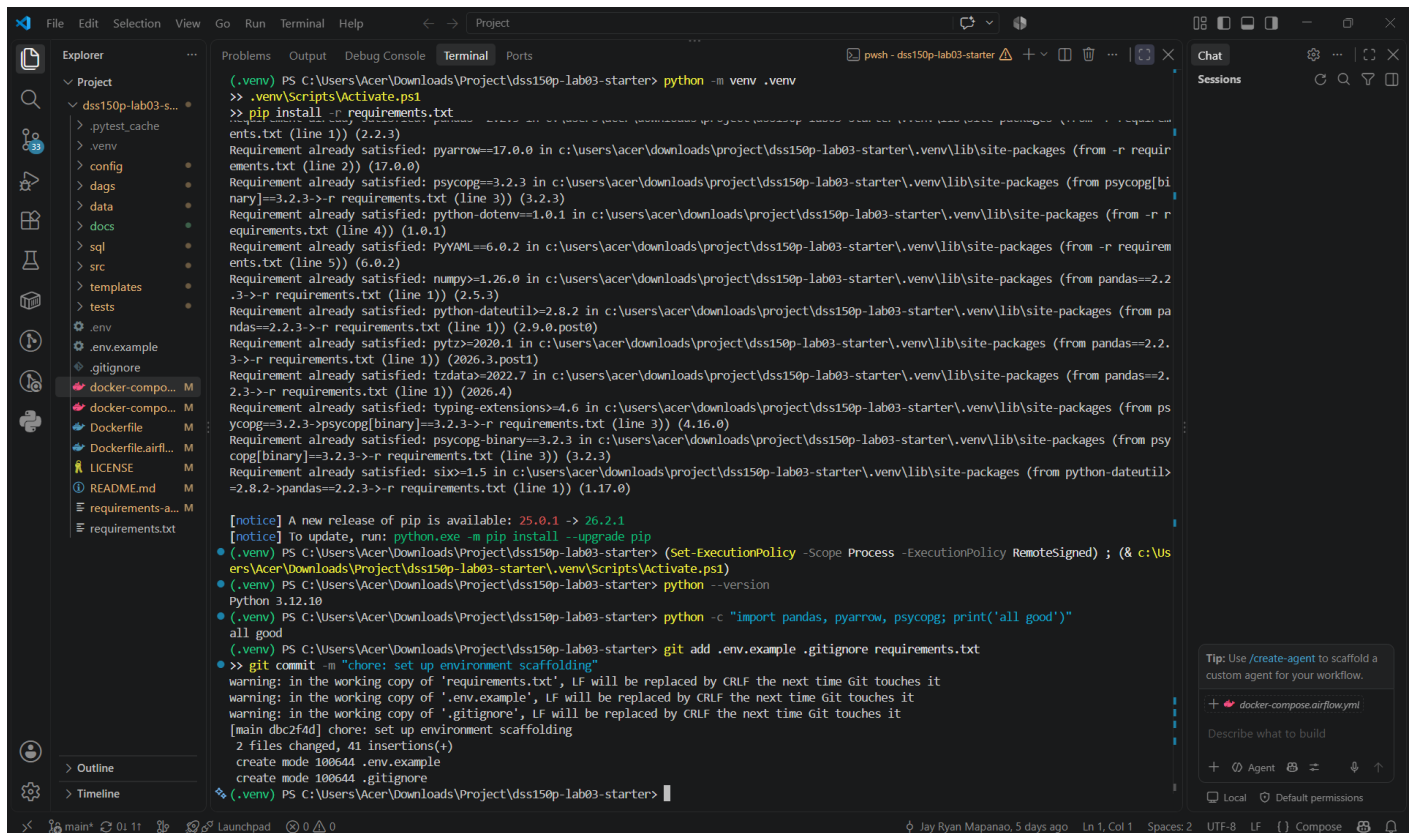
```
PS C:\Users\Acer\Downloads\Project> cd C:\Users\Acer\Downloads\Project\dss150p-lab03-starter
>> Remove-Item -Recurse -Force .venv
>> py -3.12 -m venv .venv
>> .venv\Scripts\Activate.ps1
>> pip install -r requirements.txt

[notice] A new release of pip is available: 25.0.1 -> 26.2.1
[notice] To update, run: python.exe -m pip install --upgrade pip
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> python -m venv .venv
>> .venv\Scripts\Activate.ps1
>> pip install -r requirements.txt
Error: [Errno 13] Permission denied: 'C:\Users\Acer\Downloads\Project\dss150p-lab03-starter\.venv\lib\python.exe'
Requirement already satisfied: pandas==2.2.3 in c:\users\acer\downloads\project\dss150p-lab03-starter\.venv\lib\site-packages (from -r requirements.txt (line 1)) (2.2.3)
Requirement already satisfied: pyarrow==17.0.0 in c:\users\acer\downloads\project\dss150p-lab03-starter\.venv\lib\site-packages (from -r requirements.txt (line 2)) (17.0.0)
Requirement already satisfied: pycogp==3.2.3 in c:\users\acer\downloads\project\dss150p-lab03-starter\.venv\lib\site-packages (from pycogp[binary]==3.2.3->-r requirements.txt (line 3)) (3.2.3)
Requirement already satisfied: python-dotenv==1.0.1 in c:\users\acer\downloads\project\dss150p-lab03-starter\.venv\lib\site-packages (from -r requirements.txt (line 4)) (1.0.1)
Requirement already satisfied: PyYAML==6.0.2 in c:\users\acer\downloads\project\dss150p-lab03-starter\.venv\lib\site-packages (from -r requirements.txt (line 5)) (6.0.2)
Requirement already satisfied: numpy==1.26.0 in c:\users\acer\downloads\project\dss150p-lab03-starter\.venv\lib\site-packages (from pandas==2.2.3->-r requirements.txt (line 1)) (2.5.3)
Requirement already satisfied: python-dateutil==2.8.2 in c:\users\acer\downloads\project\dss150p-lab03-starter\.venv\lib\site-packages (from pandas==2.2.3->-r requirements.txt (line 1)) (2.9.0.post0)
Requirement already satisfied: pytz==2020.1 in c:\users\acer\downloads\project\dss150p-lab03-starter\.venv\lib\site-packages (from pandas==2.2.3->-r requirements.txt (line 1)) (2026.3.post1)
Requirement already satisfied: tzdata==2022.7 in c:\users\acer\downloads\project\dss150p-lab03-starter\.venv\lib\site-packages (from pandas==2.2.3->-r requirements.txt (line 1)) (2026.4)
Requirement already satisfied: typing-extensions==4.6 in c:\users\acer\downloads\project\dss150p-lab03-starter\.venv\lib\site-packages (from pycogp==3.2.3->pycogp[binary]==3.2.3->-r requirements.txt (line 3)) (4.16.0)
Requirement already satisfied: pycogp-binary==3.2.3 in c:\users\acer\downloads\project\dss150p-lab03-starter\.venv\lib\site-packages (from pycogp[binary]==3.2.3->-r requirements.txt (line 3)) (3.2.3)
Requirement already satisfied: six==1.5 in c:\users\acer\downloads\project\dss150p-lab03-starter\.venv\lib\site-packages (from python-dateutil==2.8.2->pandas==2.2.3->-r requirements.txt (line 1)) (1.17.0)

[notice] A new release of pip is available: 25.0.1 -> 26.2.1
[notice] To update, run: python.exe -m pip install --upgrade pip
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> (Set-ExecutionPolicy -Scope Process -ExecutionPolicy RemoteSigned); (& c:\Users\Acer\Downloads\Project\dss150p-lab03-starter\.venv\Scripts\Activate.ps1)
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> python --version
Python 3.12.10
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> python -c "import pandas, pyarrow, pycogp; print('all good!')"
all good
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter>
```

The Explorer pane on the left shows the project structure, including files like `requirements.txt`, `docker-compose.yml`, and `README.md`. The Chat pane on the right contains a tip: "Tip: Use /create-agent to scaffold a custom agent for your workflow."

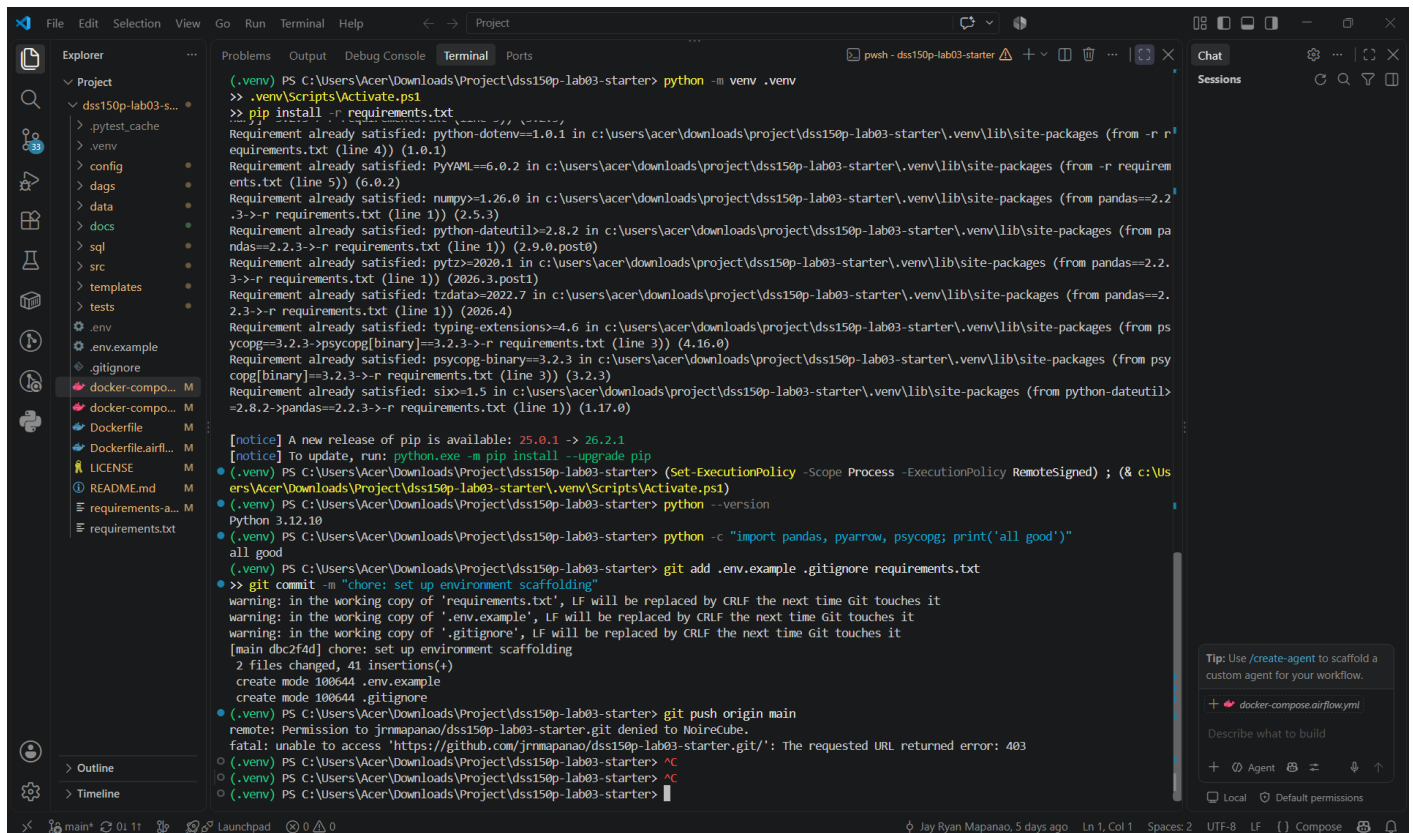
Checkpoint 1 - Confirming repo ownership (fork question)



```
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> python -m venv .venv
>> .venv\Scripts\Activate.ps1
>> pip install -r requirements.txt
Requirement already satisfied: pyarrow==17.0.0 in c:\users\acer\downloads\project\dss150p-lab03-starter\.venv\lib\site-packages (from -r requirements.txt (line 1)) (2.2.3)
Requirement already satisfied: pyarrow==17.0.0 in c:\users\acer\downloads\project\dss150p-lab03-starter\.venv\lib\site-packages (from -r requirements.txt (line 2)) (17.0.0)
Requirement already satisfied: psycogp==3.2.3 in c:\users\acer\downloads\project\dss150p-lab03-starter\.venv\lib\site-packages (from psycogp[binary]==3.2.3->-r requirements.txt (line 3)) (3.2.3)
Requirement already satisfied: python-dotenv==1.0.1 in c:\users\acer\downloads\project\dss150p-lab03-starter\.venv\lib\site-packages (from -r requirements.txt (line 4)) (1.0.1)
Requirement already satisfied: PyYAML==6.0.2 in c:\users\acer\downloads\project\dss150p-lab03-starter\.venv\lib\site-packages (from -r requirements.txt (line 5)) (6.0.2)
Requirement already satisfied: numpy==1.26.0 in c:\users\acer\downloads\project\dss150p-lab03-starter\.venv\lib\site-packages (from pandas==2.2.3->-r requirements.txt (line 1)) (2.5.3)
Requirement already satisfied: python-dateutil==2.8.2 in c:\users\acer\downloads\project\dss150p-lab03-starter\.venv\lib\site-packages (from pandas==2.2.3->-r requirements.txt (line 1)) (2.9.0.post0)
Requirement already satisfied: pytz==2020.1 in c:\users\acer\downloads\project\dss150p-lab03-starter\.venv\lib\site-packages (from pandas==2.2.3->-r requirements.txt (line 1)) (2026.3.post1)
Requirement already satisfied: tzdata==2022.7 in c:\users\acer\downloads\project\dss150p-lab03-starter\.venv\lib\site-packages (from pandas==2.2.3->-r requirements.txt (line 1)) (2026.4)
Requirement already satisfied: typing-extensions==4.6 in c:\users\acer\downloads\project\dss150p-lab03-starter\.venv\lib\site-packages (from psycogp==3.2.3->psycogp[binary]==3.2.3->-r requirements.txt (line 3)) (4.16.0)
Requirement already satisfied: psycogp-binary==3.2.3 in c:\users\acer\downloads\project\dss150p-lab03-starter\.venv\lib\site-packages (from psycogp[binary]==3.2.3->-r requirements.txt (line 3)) (3.2.3)
Requirement already satisfied: six==1.5 in c:\users\acer\downloads\project\dss150p-lab03-starter\.venv\lib\site-packages (from python-dateutil==2.8.2->pandas==2.2.3->-r requirements.txt (line 1)) (1.17.0)

[notice] A new release of pip is available: 25.0.1 -> 26.2.1
[notice] To update, run: python.exe -m pip install --upgrade pip
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> (Set-ExecutionPolicy -Scope Process -ExecutionPolicy RemoteSigned) ; (& c:\Users\Acer\Downloads\Project\dss150p-lab03-starter\.venv\Scripts\Activate.ps1)
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> python --version
Python 3.12.10
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> python -c "import pandas, pyarrow, psycogp; print('all good!')"
all good
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> git add .env.example .gitignore requirements.txt
>> git commit -m "chore: set up environment scaffolding"
warning: in the working copy of 'requirements.txt', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of '.env.example', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of '.gitignore', LF will be replaced by CRLF the next time Git touches it
[main dbc2f4d] chore: set up environment scaffolding
2 files changed, 41 insertions(+)
create mode 100644 .env.example
create mode 100644 .gitignore
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter>
```

Checkpoint 1 - git push permission denied (wrong remote)



The screenshot shows a VS Code editor with a terminal window open. The terminal displays the following commands and output:

```
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> python -m venv .venv
>> .venv\Scripts\Activate.ps1
>> pip install -r requirements.txt
Requirement already satisfied: python-dotenv==1.0.1 in c:\users\acer\downloads\project\dss150p-lab03-starter\.venv\lib\site-packages (from -r requirements.txt (line 4)) (1.0.1)
Requirement already satisfied: PyYAML==6.0.2 in c:\users\acer\downloads\project\dss150p-lab03-starter\.venv\lib\site-packages (from -r requirements.txt (line 5)) (6.0.2)
Requirement already satisfied: numpy>=1.26.0 in c:\users\acer\downloads\project\dss150p-lab03-starter\.venv\lib\site-packages (from pandas==2.2.3->-r requirements.txt (line 1)) (2.5.3)
Requirement already satisfied: python-dateutil>=2.8.2 in c:\users\acer\downloads\project\dss150p-lab03-starter\.venv\lib\site-packages (from pandas==2.2.3->-r requirements.txt (line 1)) (2.0.0.post0)
Requirement already satisfied: pytz>=2020.1 in c:\users\acer\downloads\project\dss150p-lab03-starter\.venv\lib\site-packages (from pandas==2.2.3->-r requirements.txt (line 1)) (2026.3.post1)
Requirement already satisfied: tzdata>=2022.7 in c:\users\acer\downloads\project\dss150p-lab03-starter\.venv\lib\site-packages (from pandas==2.2.3->-r requirements.txt (line 1)) (2026.4)
Requirement already satisfied: typing-extensions>=4.6 in c:\users\acer\downloads\project\dss150p-lab03-starter\.venv\lib\site-packages (from psycopg==3.2.3->-r requirements.txt (line 3)) (4.16.0)
Requirement already satisfied: psycopg-binary==3.2.3 in c:\users\acer\downloads\project\dss150p-lab03-starter\.venv\lib\site-packages (from psycopg==3.2.3->-r requirements.txt (line 3)) (3.2.3)
Requirement already satisfied: six>=1.5 in c:\users\acer\downloads\project\dss150p-lab03-starter\.venv\lib\site-packages (from python-dateutil==2.8.2->pandas==2.2.3->-r requirements.txt (line 1)) (1.17.0)

[notice] A new release of pip is available: 25.0.1 -> 26.2.1
[notice] To update, run: python.exe -m pip install --upgrade pip
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> (Set-ExecutionPolicy -Scope Process -ExecutionPolicy RemoteSigned) ; (& c:\Users\Acer\Downloads\Project\dss150p-lab03-starter\.venv\Scripts\Activate.ps1)
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> python --version
Python 3.12.10
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> python -c "import pandas, pyarrow, psycopg; print('all good!')"
all good
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> git add .env.example .gitignore requirements.txt
>> git commit -m "chore: set up environment scaffolding"
warning: in the working copy of 'requirements.txt', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of '.env.example', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of '.gitignore', LF will be replaced by CRLF the next time Git touches it
[main dbc2f4d] chore: set up environment scaffolding
2 files changed, 41 insertions(+)
create mode 100644 .env.example
create mode 100644 .gitignore
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> git push origin main
remote: Permission to jrmpanao/dss150p-lab03-starter.git denied to NoireCube.
fatal: unable to access 'https://github.com/jrmpanao/dss150p-lab03-starter.git/': The requested URL returned error: 403
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> ^C
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> ^C
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> ^C
```

The error message indicates that the user does not have permission to push to the repository. The error is: "remote: Permission to jrmpanao/dss150p-lab03-starter.git denied to NoireCube. fatal: unable to access 'https://github.com/jrmpanao/dss150p-lab03-starter.git/': The requested URL returned error: 403".

Checkpoint 1 - Setting remote to personal fork

The screenshot shows a Windows terminal window with the following content:

```

(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> python -m venv .venv
>> .venv\Scripts\Activate.ps1
>> pip install -r requirements.txt
Requirement already satisfied: PyYAML==6.0.2 in c:\users\acer\downloads\project\dss150p-lab03-starter\.venv\lib\site-packages (from -r requirements.txt (line 5)) (6.0.2)
Requirement already satisfied: numpy>=1.26.0 in c:\users\acer\downloads\project\dss150p-lab03-starter\.venv\lib\site-packages (from pandas==2.2.3->-r requirements.txt (line 1)) (2.5.3)
Requirement already satisfied: python-dateutil>=2.8.2 in c:\users\acer\downloads\project\dss150p-lab03-starter\.venv\lib\site-packages (from pandas==2.2.3->-r requirements.txt (line 1)) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in c:\users\acer\downloads\project\dss150p-lab03-starter\.venv\lib\site-packages (from pandas==2.2.3->-r requirements.txt (line 1)) (2026.3.post1)
Requirement already satisfied: tzdata>=2022.7 in c:\users\acer\downloads\project\dss150p-lab03-starter\.venv\lib\site-packages (from pandas==2.2.3->-r requirements.txt (line 1)) (2026.4)
Requirement already satisfied: typing-extensions>=4.6 in c:\users\acer\downloads\project\dss150p-lab03-starter\.venv\lib\site-packages (from psycopy==3.2.3->psycopy[binary]==3.2.3->-r requirements.txt (line 3)) (4.16.0)
Requirement already satisfied: psycopy-binary==3.2.3 in c:\users\acer\downloads\project\dss150p-lab03-starter\.venv\lib\site-packages (from psycopy[binary]==3.2.3->-r requirements.txt (line 3)) (3.2.3)
Requirement already satisfied: six>=1.5 in c:\users\acer\downloads\project\dss150p-lab03-starter\.venv\lib\site-packages (from python-dateutil>=2.8.2->pandas==2.2.3->-r requirements.txt (line 1)) (1.17.0)

[notice] A new release of pip is available: 25.0.1 -> 26.2.1
[notice] To update, run: python.exe -m pip install --upgrade pip
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> git init
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> python -v
Python 3.12.10
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> python -c "import pandas, pyarrow, psycopy; print('all good!')"
all good
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> git add .env.example .gitignore requirements.txt
>> git commit -m "chore: set up environment scaffolding"
warning: in the working copy of 'requirements.txt', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of '.env.example', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of '.gitignore', LF will be replaced by CRLF the next time Git touches it
[main dbc2f4d] chore: set up environment scaffolding
 2 files changed, 41 insertions(+)
 create mode 100644 .env.example
 create mode 100644 .gitignore
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> git push origin main
remote: Permission to jrmapanao/dss150p-lab03-starter.git denied to Noirecube.
fatal: unable to access 'https://github.com/jrmapanao/dss150p-lab03-starter.git/': The requested URL returned error: 403
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter>
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> git remote set-url origin https://github.com/Noirecube/dss150p-lab03-starter.git
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter>
  
```

Checkpoint 1 - git push succeeded to fork

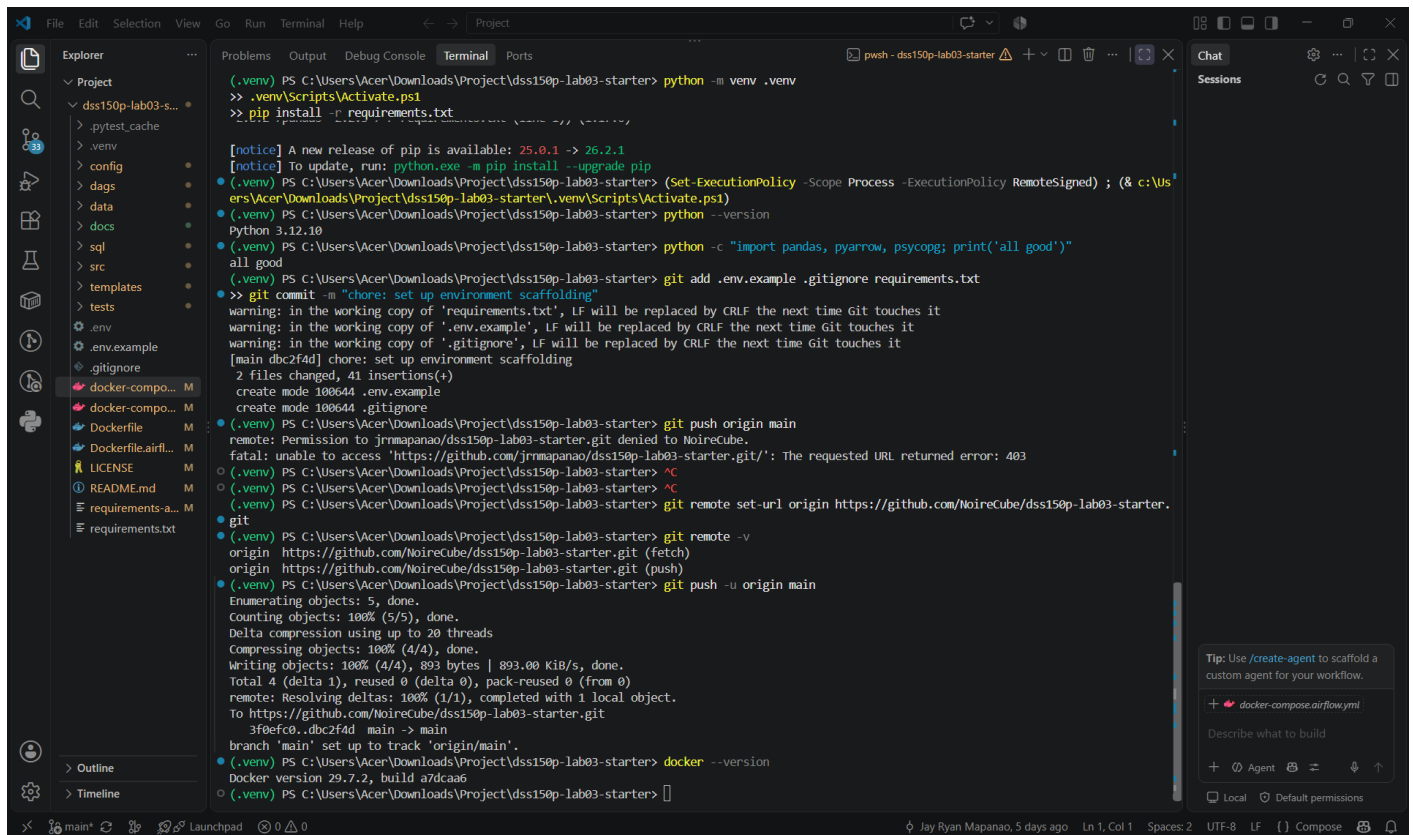
The screenshot shows a VS Code terminal window with the following commands and output:

```
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> python -m venv .venv
>> .venv\Scripts\Activate.ps1
>> pip install -r requirements.txt
Requirement already satisfied: six>=1.5 in c:\users\acer\downloads\project\dss150p-lab03-starter\.venv\lib\site-packages (from python-dateutil
=2.8.2->pandas==2.2.3->-r requirements.txt (line 1)) (1.17.0)

[notice] A new release of pip is available: 25.0.1 -> 26.2.1
[notice] To update, run: python.exe -m pip install --upgrade pip
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> (Set-ExecutionPolicy -Scope Process -ExecutionPolicy RemoteSigned) ; (& c:\Us
ers\Acer\Downloads\Project\dss150p-lab03-starter\.venv\Scripts\Activate.ps1)
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> python --version
Python 3.12.10
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> python -c "import pandas, pyarrow, psycog; print('all good!')"
all good
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> git add .env.example .gitignore requirements.txt
>> git commit -m "chore: set up environment scaffolding"
warning: in the working copy of 'requirements.txt', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of '.env.example', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of '.gitignore', LF will be replaced by CRLF the next time Git touches it
[main dbc2fad] chore: set up environment scaffolding
2 files changed, 41 insertions(+)
create mode 100644 .env.example
create mode 100644 .gitignore
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> git push origin main
remote: Permission to jrmapanao/dss150p-lab03-starter.git denied to NoireCube.
fatal: unable to access 'https://github.com/jrmapanao/dss150p-lab03-starter.git/': The requested URL returned error: 403
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> ^C
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> ^C
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> git remote set-url origin https://github.com/NoireCube/dss150p-lab03-starter.
git
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> git remote -v
origin https://github.com/NoireCube/dss150p-lab03-starter.git (fetch)
origin https://github.com/NoireCube/dss150p-lab03-starter.git (push)
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> git push -u origin main
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 20 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (4/4), 893 bytes | 893.00 KiB/s, done.
Total 4 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/NoireCube/dss150p-lab03-starter.git
3f0efc0..dbc2fad main -> main
branch 'main' set up to track 'origin/main'.
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter>
```

The terminal output shows the successful completion of the git push command, indicating that the code has been pushed to the forked repository.

Checkpoint 1 - Docker version confirmed installed



```
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> python -m venv .venv
>> .venv\Scripts\Activate.ps1
>> pip install -r requirements.txt

[notice] A new release of pip is available: 25.0.1 -> 26.2.1
[notice] To update, run: python.exe -m pip install --upgrade pip
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> (Set-ExecutionPolicy -Scope Process -ExecutionPolicy RemoteSigned) ; (& c:\Us
ers\Acer\Downloads\Project\dss150p-lab03-starter\.venv\Scripts\Activate.ps1)
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> python --version
Python 3.12.10
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> python -c "import pandas, pyarrow, psycopg; print('all good!')"
all good
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> git add .env.example .gitignore requirements.txt
>> git commit -m "chore: set up environment scaffolding"
warning: in the working copy of 'requirements.txt', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of '.env.example', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of '.gitignore', LF will be replaced by CRLF the next time Git touches it
[main dbc2f4d] chore: set up environment scaffolding
2 files changed, 41 insertions(+)
create mode 100644 .env.example
create mode 100644 .gitignore
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> git push origin main
remote: Permission to jrmapanao/dss150p-lab03-starter.git denied to NoireCube.
fatal: unable to access 'https://github.com/jrmapanao/dss150p-lab03-starter.git/': The requested URL returned error: 403
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> ^C
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> ^C
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> git remote set-url origin https://github.com/NoireCube/dss150p-lab03-starter.
git
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> git remote -v
origin https://github.com/NoireCube/dss150p-lab03-starter.git (fetch)
origin https://github.com/NoireCube/dss150p-lab03-starter.git (push)
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> git push -u origin main
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 20 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (4/4), 893 bytes | 893.00 KiB/s, done.
Total 4 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/NoireCube/dss150p-lab03-starter.git
3f0efc0..dbc2f4d main -> main
branch 'main' set up to track 'origin/main'.
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> docker --version
Docker version 29.7.2, build a7dca6e
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> ^
```

PostgreSQL setup - container name conflict (leftover container)

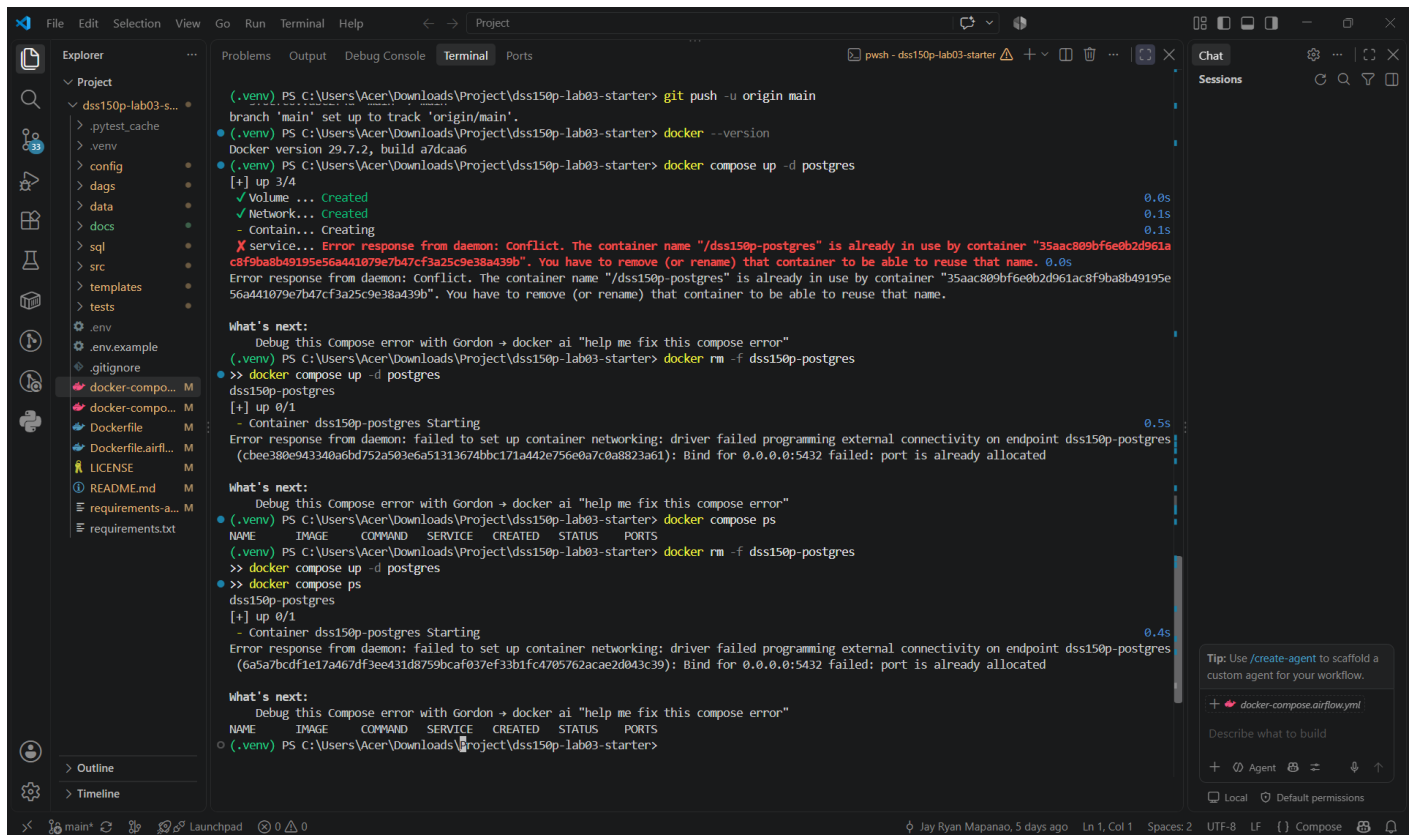
```

(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> git add .env.example .gitignore requirements.txt
>> git commit -m "chore: set up environment scaffolding"
warning: in the working copy of 'requirements.txt', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of '.env.example', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of '.gitignore', LF will be replaced by CRLF the next time Git touches it
[main dbc2fad] chore: set up environment scaffolding
2 files changed, 41 insertions(+)
create mode 100644 .env.example
create mode 100644 .gitignore
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> git push origin main
remote: Permission to jrmapanao/dss150p-lab03-starter.git denied to NoireCube.
fatal: unable to access 'https://github.com/jrmapanao/dss150p-lab03-starter.git/': The requested URL returned error: 403
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> ^C
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> ^C
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> git remote set-url origin https://github.com/NoireCube/dss150p-lab03-starter.git
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> git remote -v
origin https://github.com/NoireCube/dss150p-lab03-starter.git (fetch)
origin https://github.com/NoireCube/dss150p-lab03-starter.git (push)
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> git push -u origin main
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 20 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (4/4), 893 bytes | 893.00 KiB/s, done.
Total 4 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/NoireCube/dss150p-lab03-starter.git
3f0efc0..dbc2fad main -> main
branch 'main' set up to track 'origin/main'.
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> docker --version
Docker version 29.7.2, build a7dcaae
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> docker compose up -d postgres
[+] up 3/4
  ✓ Volume ... Created                                0.0s
  ✓ Network... Created                                0.1s
  - Contain... Creating                                0.1s
  ✗ service... Error response from daemon: Conflict. The container name "/dss150p-postgres" is already in use by container "35aac809bf6e0b2d961ac8f9ba8b49195e56a441079e7b47cf3a25c9e38a439b". You have to remove (or rename) that container to be able to reuse that name. 0.0s
Error response from daemon: Conflict. The container name "/dss150p-postgres" is already in use by container "35aac809bf6e0b2d961ac8f9ba8b49195e56a441079e7b47cf3a25c9e38a439b". You have to remove (or rename) that container to be able to reuse that name.

What's next:
  Debug this Compose error with Gordon → docker ai "help me fix this compose error"
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter>

```

PostgreSQL setup - port 5432 already allocated error



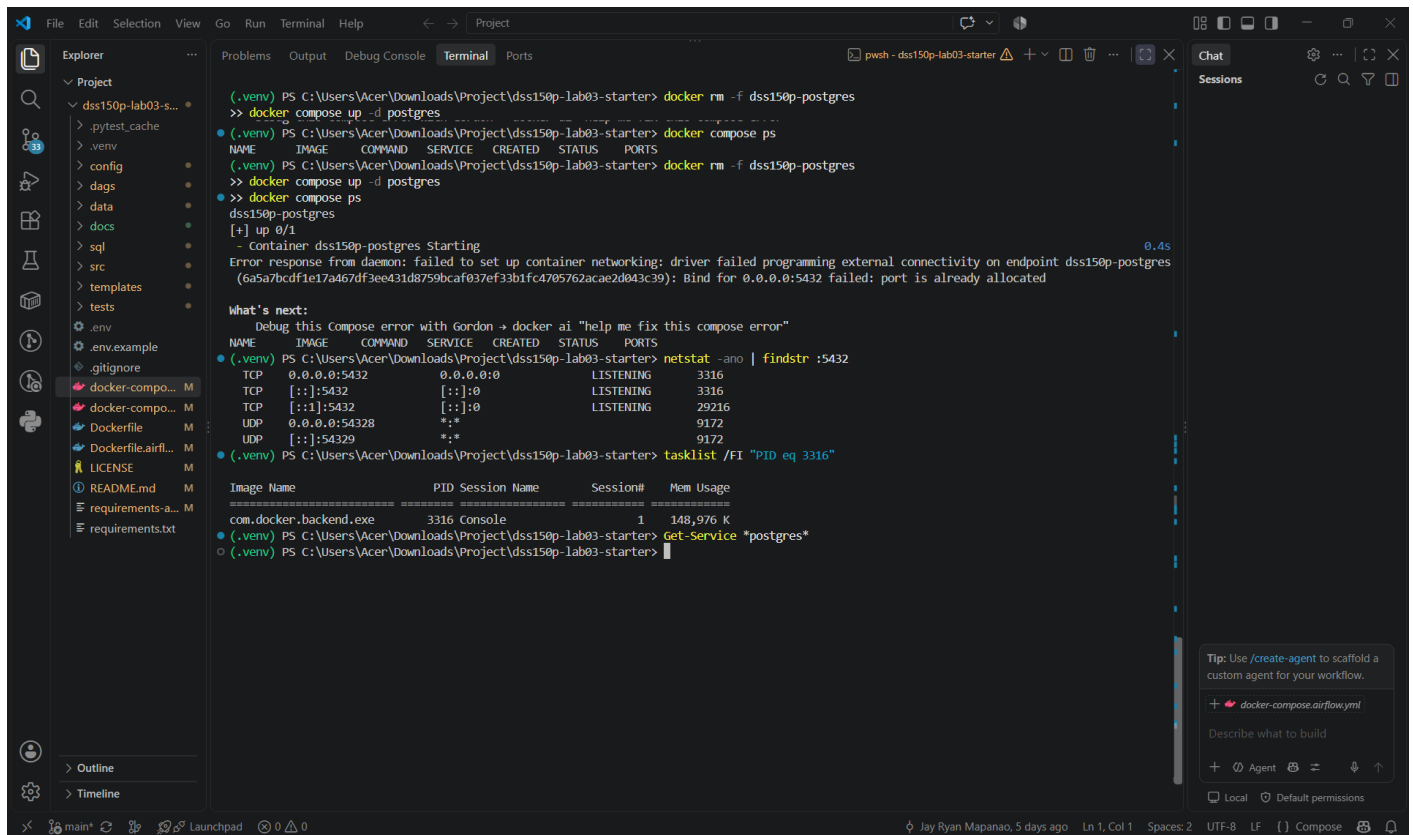
```
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> git push -u origin main
branch 'main' set up to track 'origin/main'.
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> docker --version
Docker version 29.7.2, build a7dca6
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> docker compose up -d postgres
[+] up 3/4
  ✓ Volume ... Created                                0.0s
  ✓ Network ... Created                                0.1s
  - Container ... Creating                             0.1s
  X service... Error response from daemon: Conflict. The container name "/dss150p-postgres" is already in use by container "35aac809bf6e0b2d961ac8f9ba8b49195e56a441079e7b47cf3a25c9e38a439b". You have to remove (or rename) that container to be able to reuse that name. 0.0s
Error response from daemon: Conflict. The container name "/dss150p-postgres" is already in use by container "35aac809bf6e0b2d961ac8f9ba8b49195e56a441079e7b47cf3a25c9e38a439b". You have to remove (or rename) that container to be able to reuse that name.

What's next:
  Debug this Compose error with Gordon → docker ai "help me fix this compose error"
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> docker rm -f dss150p-postgres
dss150p-postgres
[+] up 0/1
  - Container dss150p-postgres Starting
Error response from daemon: failed to set up container networking: driver failed programming external connectivity on endpoint dss150p-postgres (cbee380e943340a0bd752a503e6a51313674bbc171a442e756e0a7c0a8823a61): Bind for 0.0.0.0:5432 failed: port is already allocated
0.5s

What's next:
  Debug this Compose error with Gordon → docker ai "help me fix this compose error"
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> docker compose ps
NAME          IMAGE          COMMAND          SERVICE          CREATED          STATUS          PORTS
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> docker rm -f dss150p-postgres
dss150p-postgres
[+] up 0/1
  - Container dss150p-postgres Starting
Error response from daemon: failed to set up container networking: driver failed programming external connectivity on endpoint dss150p-postgres (6a5a78cdf1e17a467df3ee431d8759bcacf037ef33b1fc4705762acae2d043c39): Bind for 0.0.0.0:5432 failed: port is already allocated
0.4s

What's next:
  Debug this Compose error with Gordon → docker ai "help me fix this compose error"
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter>
```

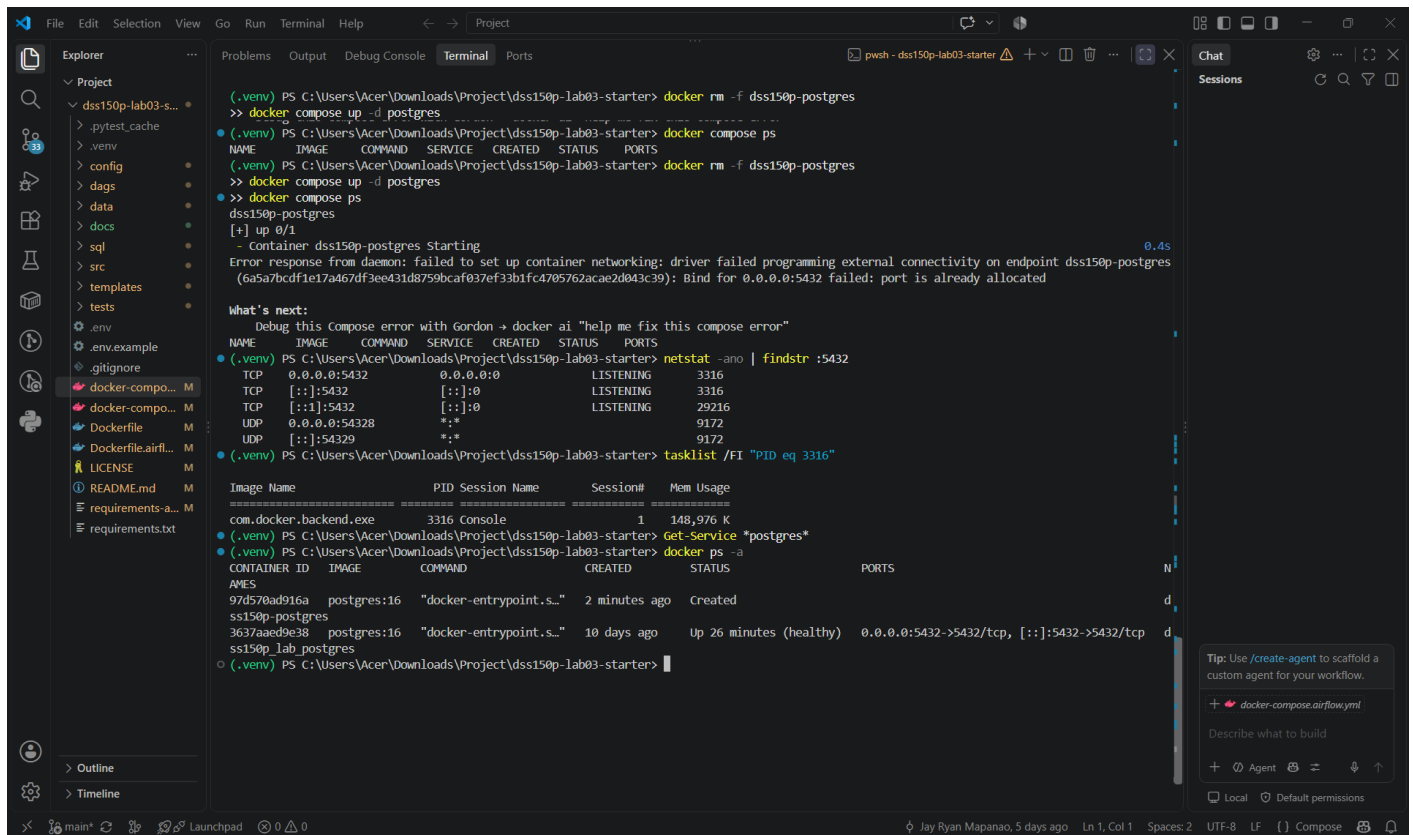
PostgreSQL setup - identifying process holding port 5432



```
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> docker rm -f dss150p-postgres
>> docker compose up -d postgres
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> docker compose ps
NAME                IMAGE                COMMAND              SERVICE   CREATED   STATUS    PORTS
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> docker rm -f dss150p-postgres
>> docker compose up -d postgres
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> docker compose ps
dss150p-postgres
[+] up 0/1
- Container dss150p-postgres Starting
Error response from daemon: failed to set up container networking: driver failed programming external connectivity on endpoint dss150p-postgres
(6a5a7bcd1e17a467df3ee431d8759bcacf037ef33b1fc4705762acae2d043c39): Bind for 0.0.0.0:5432 failed: port is already allocated

What's next:
  Debug this Compose error with Gordon → docker ai "help me fix this compose error"
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> netstat -ano | findstr :5432
TCP        0.0.0.0:5432          0.0.0.0:0          LISTENING   3316
TCP        [::]:5432            [::]:0             LISTENING   3316
TCP        [::1]:5432           [::1]:0            LISTENING   29216
UDP        0.0.0.0:54328        *:~                LISTENING   9172
UDP        [::]:54329           *:~                LISTENING   9172
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> tasklist /FI "PID eq 3316"
Image Name                PID Session Name        Session#    Mem Usage
=====
com.docker.backend.exe     3316 Console              1           148,976 K
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> Get-Service *postgres*
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter>
```

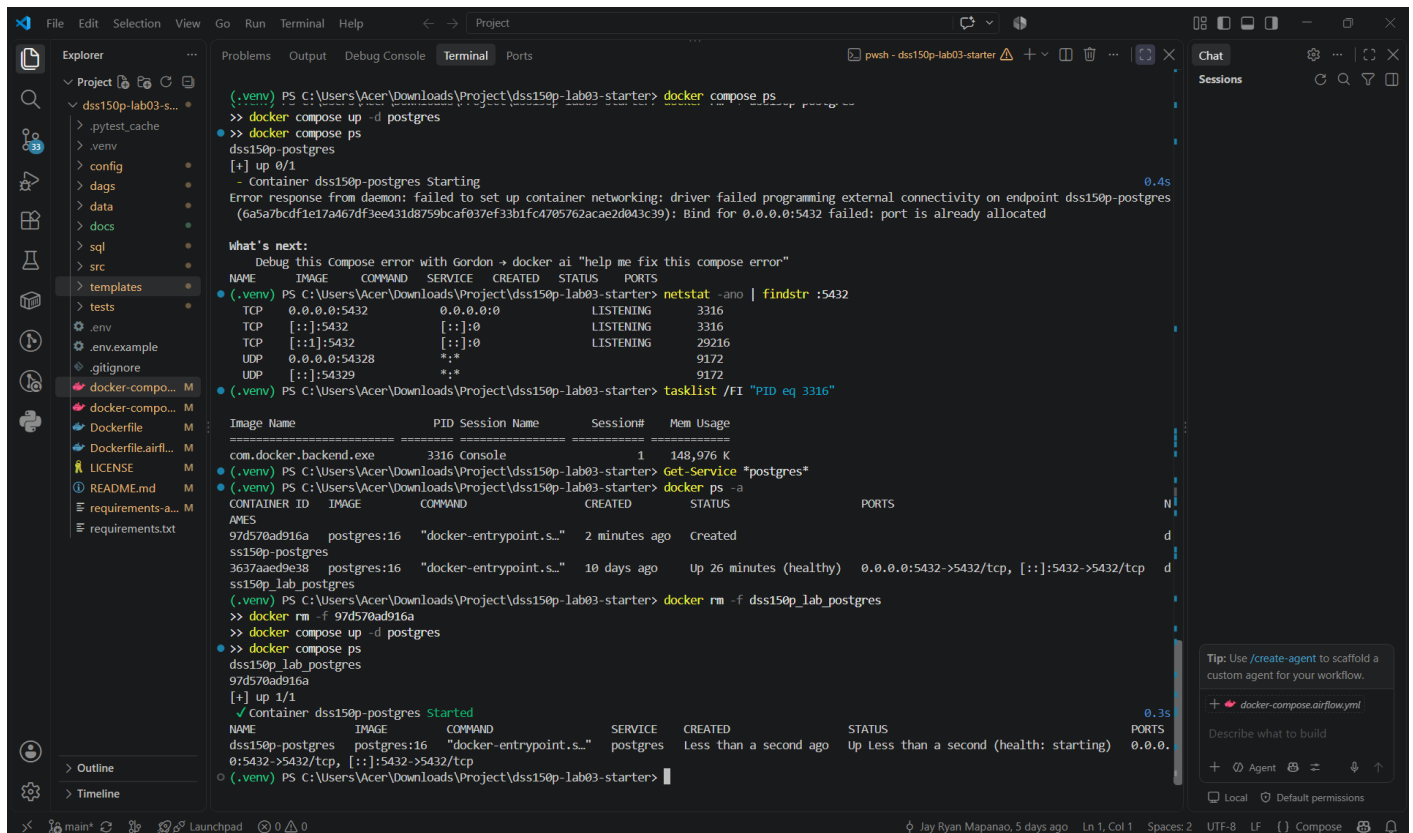
PostgreSQL setup - found conflicting old container (docker ps -a)



```
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> docker rm -f dss150p-postgres
>> docker compose up -d postgres
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> docker compose ps
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> docker rm -f dss150p-postgres
>> docker compose up -d postgres
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> docker ps
dss150p-postgres
[+] up 0/1
- Container dss150p-postgres Starting
Error response from daemon: failed to set up container networking: driver failed programming external connectivity on endpoint dss150p-postgres
(6a5a7bcd1e17a467df3ee431d8759bcdf037ef33b1fc4705762acae2d043c39): Bind for 0.0.0.0:5432 failed: port is already allocated

What's next:
  Debug this Compose error with Gordon → docker ai "help me fix this compose error"
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> netstat -ano | findstr :5432
TCP        0.0.0.0:5432          0.0.0.0:0           LISTENING   3316
TCP        [::]:5432            [::]:0              LISTENING   3316
TCP        [::1]:5432           [::1]:0              LISTENING   29216
UDP        0.0.0.0:54328        *:*                  LISTENING   9172
UDP        [::]:54329          *:*                  LISTENING   9172
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> tasklist /FI "PID eq 3316"
Image Name      PID Session Name  Session#  Mem Usage
-----
com.docker.backend.exe 3316 Console      1         148,976 K
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> Get-Service "postgres"
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> docker ps -a
CONTAINER ID   IMAGE      COMMAND                  CREATED        STATUS        PORTS
AMES          postgres:16 "docker-entrypoint.s..." 2 minutes ago  Created
dss150p-postgres
3637aaed9e38   postgres:16 "docker-entrypoint.s..." 10 days ago   Up 26 minutes (healthy)   0.0.0.0:5432->5432/tcp, [::]:5432->5432/tcp
dss150p_lab_postgres
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter>
```

PostgreSQL setup - Postgres container started successfully



```
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> docker compose ps
>> docker compose up -d postgres
>> docker compose ps
dss150p-postgres
[+] up 0/1
- Container dss150p-postgres Starting
Error response from daemon: failed to set up container networking: driver failed programming external connectivity on endpoint dss150p-postgres
(6a5a7bcd1e17a467df3ee431d8759bcaf037ef33b1fc4705762acae2d043c39): Bind for 0.0.0.0:5432 failed: port is already allocated

What's next:
Debug this Compose error with Gordon -> docker ai "help me fix this compose error"

(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> netstat -ano | findstr :5432
TCP 0.0.0.0:5432 0.0.0.0:0 LISTENING 3316
TCP [::]:5432 [::]:0 LISTENING 3316
TCP [::]:5432 [::]:0 LISTENING 29216
UDP 0.0.0.0:54328 *:* 9172
UDP [::]:54329 *:* 9172

(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> tasklist /FI "PID eq 3316"

Image Name PID Session Name Session# Mem Usage
-----
com.docker.backend.exe 3316 Console 1 148,976 K

(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> Get-Service *postgres*
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> docker ps -a
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS
AMES
97d570ad916a postgres:16 "docker-entrypoint.s..." 2 minutes ago Created
ss150p-postgres
3637aaed9e38 postgres:16 "docker-entrypoint.s..." 10 days ago Up 26 minutes (healthy) 0.0.0.0:5432->5432/tcp, [::]:5432->5432/tcp d
ss150p_lab_postgres
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> docker rm -f dss150p_lab_postgres
>> docker rm -f 97d570ad916a
>> docker compose up -d postgres
>> docker compose ps
dss150p_lab_postgres
97d570ad916a
[+] up 1/1
✓ Container dss150p-postgres Started

NAME IMAGE COMMAND SERVICE CREATED STATUS PORTS
dss150p-postgres postgres:16 "docker-entrypoint.s..." postgres Less than a second ago Up Less than a second (health: starting) 0.0.0.0:5432->5432/tcp, [::]:5432->5432/tcp
```

PostgreSQL setup - warehouse schema applied

```
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> docker rm -f dss150p-postgres
>> docker compose up -d postgres
>> docker compose ps
TCP 0.0.0.0:5432 0.0.0.0:0 LISTENING 3316
TCP [::]:5432 [::]:0 LISTENING 3316
TCP [::1]:5432 [::]:0 LISTENING 29216
UDP 0.0.0.0:54328 *:~ 9172
UDP [::]:54329 *:~ 9172
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> tasklist /FI "PID eq 3316"
Image Name PID Session Name Session# Mem Usage
-----
com.docker.backend.exe 3316 Console 1 148,976 K
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> Get-Service "postgres"
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> docker ps -a
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS NAMES
97d570ad916a postgres:16 "docker-entrypoint.s..." 2 minutes ago Created dss150p-postgres
36372aed9e38 postgres:16 "docker-entrypoint.s..." 10 days ago Up 26 minutes (healthy) 0.0.0.0:5432->5432/tcp, [::]:5432->5432/tcp dss150p_lab_postgres
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> docker rm -f dss150p_lab_postgres
>> docker rm -f 97d570ad916a
>> docker compose up -d postgres
>> docker compose ps
NAME IMAGE COMMAND SERVICE CREATED STATUS PORTS
dss150p-postgres postgres:16 "docker-entrypoint.s..." postgres Less than a second ago Up Less than a second (health: starting) 0.0.0.0:5432->5432/tcp, [::]:5432->5432/tcp
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> docker exec -i dss150p-postgres psql -U dss150p -d dss150p < sql\init\01_ware
house_schema.sql
ParserError:
Line
1 | docker exec -i dss150p-postgres psql -U dss150p -d dss150p < sql\init ...
|
| The '<' operator is reserved for future use.
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter>
```

Tip: Use `/create-agent` to scaffold a custom agent for your workflow.

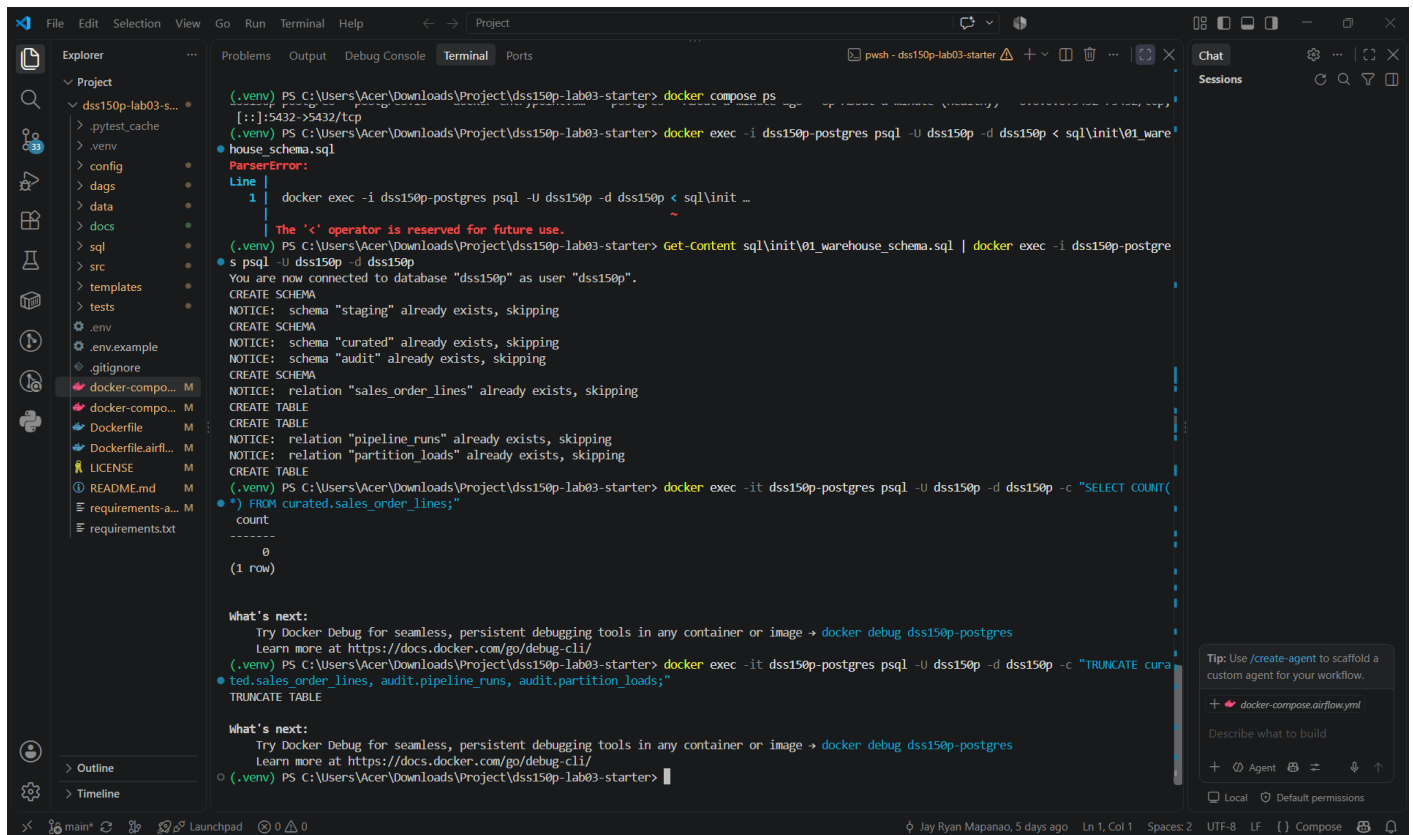
+ `docker-compose.airflow.yml`

Describe what to build

+ Agent

Local Default permissions

PostgreSQL setup - verifying clean database (truncate check)



```
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> docker compose ps
[::]:5432->5432/tcp
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> docker exec -i dss150p-postgres psql -U dss150p -d dss150p < sql\init\01_ware
house_schema.sql
ParserError:
Line
1 | docker exec -i dss150p-postgres psql -U dss150p -d dss150p < sql\init ...
~
The '<' operator is reserved for future use.
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> Get-Content sql\init\01_warehouse_schema.sql | docker exec -i dss150p-postgre
s psql -U dss150p -d dss150p
You are now connected to database "dss150p" as user "dss150p".
CREATE SCHEMA
NOTICE: schema "staging" already exists, skipping
CREATE SCHEMA
NOTICE: schema "curated" already exists, skipping
NOTICE: schema "audit" already exists, skipping
CREATE SCHEMA
NOTICE: relation "sales_order_lines" already exists, skipping
CREATE TABLE
CREATE TABLE
NOTICE: relation "pipeline_runs" already exists, skipping
NOTICE: relation "partition_loads" already exists, skipping
CREATE TABLE
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> docker exec -it dss150p-postgres psql -U dss150p -d dss150p -c "SELECT COUNT(
*) FROM curated.sales_order_lines;"
count
-----
0
(1 row)

What's next:
Try Docker Debug for seamless, persistent debugging tools in any container or image → docker debug dss150p-postgres
Learn more at https://docs.docker.com/go/debug-cli/
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> docker exec -it dss150p-postgres psql -U dss150p -d dss150p -c "TRUNCATE cura
ted.sales_order_lines, audit.pipeline_runs, audit.partition_loads;"
TRUNCATE TABLE

What's next:
Try Docker Debug for seamless, persistent debugging tools in any container or image → docker debug dss150p-postgres
Learn more at https://docs.docker.com/go/debug-cli/
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter>
```

Checkpoint 2 - extract & transform run (staging/curated row counts)

```
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> Get-Content sql\init\01_warehouse_schema.sql | docker exec -i dss150p-postgres psql -U dss150p -d dss150p
CREATE TABLE
NOTICE: relation "pipeline_runs" already exists, skipping
NOTICE: relation "partition_loads" already exists, skipping
CREATE TABLE
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> docker exec -it dss150p-postgres psql -U dss150p -d dss150p -c "SELECT COUNT(*) FROM curated.sales_order_lines;"
count
-----
0
(1 row)

What's next:
Try Docker Debug for seamless, persistent debugging tools in any container or image → docker debug dss150p-postgres
Learn more at https://docs.docker.com/go/debug-cli/
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> docker exec -it dss150p-postgres psql -U dss150p -d dss150p -c "TRUNCATE curated.sales_order_lines, audit.pipeline_runs, audit.partition_loads;"
TRUNCATE TABLE

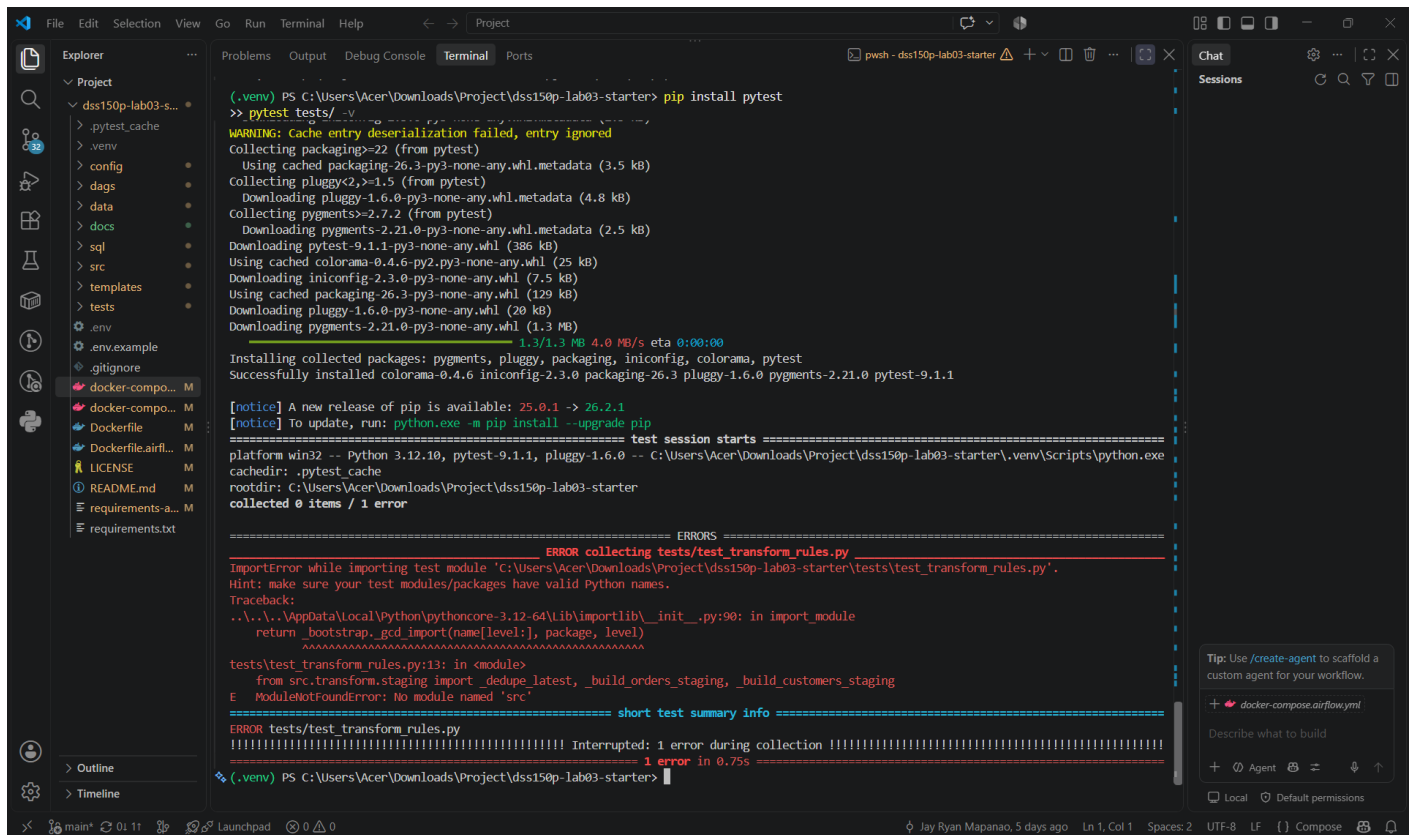
What's next:
Try Docker Debug for seamless, persistent debugging tools in any container or image → docker debug dss150p-postgres
Learn more at https://docs.docker.com/go/debug-cli/
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> python -m src.cli validate-env
PROJECT_ROOT= c:\Users\Acer\Downloads\Project\dss150p-lab03-starter
DB host/database= localhost dss150p
Configured source= data/source
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> python -m src.cli extract
>> python -m src.cli transform
run_id=run_20260923T072401Z_9ce78de4
raw_dir=C:\Users\Acer\Downloads\Project\dss150p-lab03-starter\data\raw\run_id=run_20260923T072401Z_9ce78de4
copied customers.csv
copied orders.csv
copied products.json
run_id=run_20260923T072402Z_ca734532
staging.customers rows=3000
staging.products rows=599
staging.orders rows=49998
staging.quarantine rows=3
curated rows=49897
curated.quarantine rows=101
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter>
```

Checkpoint 3 - load run x2 proving idempotency (0 rows on rerun) + validation passed

```
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> python -m src.cli extract
>> python -m src.cli transform
run_id=run_20260923T072401Z_9ce78de4
raw_dir=C:\Users\Acer\Downloads\Project\dss150p-lab03-starter\data\raw\run_id=run_20260923T072401Z_9ce78de4
copied customers.csv
copied orders.csv
copied products.json
run_id=run_20260923T072402Z_ca734532
staging.customers rows=3000
staging.products rows=599
staging.orders rows=49998
staging.quarantine rows=3
curated rows=49897
curated.quarantine rows=101
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> python -m src.cli load
run_id=run_20260923T072500Z_3cf245d8
staging.customers rows=3000
staging.products rows=599
staging.orders rows=49998
staging.quarantine rows=3
curated rows=49897
curated.quarantine rows=101
upsert_affected_rows=49897
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> python -m src.cli load
run_id=run_20260923T072516Z_09d8cb26
staging.customers rows=3000
staging.products rows=599
staging.orders rows=49998
staging.quarantine rows=3
curated rows=49897
curated.quarantine rows=101
upsert_affected_rows=0
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> docker exec -it dss150p-postgres psql -U dss150p -d dss150p -c "SELECT COUNT(*) total, COUNT(DISTINCT order_id) distinct_orders FROM curated.sales_order_lines;"
total | distinct_orders
-----|-----
49897 | 49897
(1 row)

What's next:
Try Docker Debug for seamless, persistent debugging tools in any container or image → docker debug dss150p-postgres
Learn more at https://docs.docker.com/go/debug-cli/
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter>
```

Checkpoint 4 - pytest not found (missing dependency)



```
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> pip install pytest
>> pytest tests/ -v
WARNING: Cache entry deserialization failed, entry ignored
Collecting packaging>=22 (from pytest)
  Using cached packaging-26.3-py3-none-any.whl.metadata (3.5 kB)
Collecting pluggy<2,>=1.5 (from pytest)
  Downloading pluggy-1.6.0-py3-none-any.whl.metadata (4.8 kB)
Collecting pygments>=2.7.2 (from pytest)
  Downloading pygments-2.21.0-py3-none-any.whl.metadata (2.5 kB)
  Downloading pytest-9.1.1-py3-none-any.whl (386 kB)
Using cached colorama-0.4.6-py2.py3-none-any.whl (25 kB)
  Downloading iniconfig-2.3.0-py3-none-any.whl (7.5 kB)
Using cached packaging-26.3-py3-none-any.whl (129 kB)
  Downloading pluggy-1.6.0-py3-none-any.whl (20 kB)
  Downloading pygments-2.21.0-py3-none-any.whl (1.3 MB)
1.3/1.3 MB 4.0 MB/s eta 0:00:00
Installing collected packages: pygments, pluggy, packaging, iniconfig, colorama, pytest
Successfully installed colorama-0.4.6 iniconfig-2.3.0 packaging-26.3 pluggy-1.6.0 pygments-2.21.0 pytest-9.1.1

[notice] A new release of pip is available: 25.0.1 -> 26.2.1
[notice] To update, run: python.exe -m pip install --upgrade pip
===== test session starts =====
platform win32 -- Python 3.12.10, pytest-9.1.1, pluggy-1.6.0 -- C:\Users\Acer\Downloads\Project\dss150p-lab03-starter\.venv\Scripts\python.exe
cachedir: .pytest_cache
rootdir: c:\Users\Acer\Downloads\Project\dss150p-lab03-starter
collected 0 items / 1 error

===== ERRORS =====
ERROR collecting tests/test_transform_rules.py
ImportError while importing test module 'C:\Users\Acer\Downloads\Project\dss150p-lab03-starter\tests\test_transform_rules.py'.
Hint: make sure your test modules/packages have valid Python names.
Traceback:
..\..\AppData\Local\Python\pythoncore-3.12-64\lib\importlib\_init_.py:90: in import_module
    return _bootstrap._gcd_import(name[level:], package, level)
tests\test_transform_rules.py:13: in <module>
    from src.transform.staging import _dedupe_latest, _build_orders_staging, _build_customers_staging
E   ModuleNotFoundError: No module named 'src'

===== short test summary info =====
Interrupted: 1 error during collection !!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!
===== 1 error in 0.75s =====
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter>
```

Checkpoint 4 - all 6 unit tests passed; Checkpoint 5 - benchmark results

The screenshot shows a VS Code interface with a terminal window open. The terminal displays the following commands and output:

```
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> pip install pytest
>> pytest tests/ -v
===== 1 error in 0.75s =====
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> python -m pytest tests/ -v
===== test session starts =====
platform win32 -- Python 3.12.10, pytest-9.1.1, pluggy-1.6.0 -- C:\Users\Acer\Downloads\Project\dss150p-lab03-starter\.venv\Scripts\python.exe
cachedir: .pytest_cache
rootdir: C:\Users\Acer\Downloads\Project\dss150p-lab03-starter
collected 6 items

tests/test_transform_rules.py::test dedupe latest keeps greatest updated at PASSED [ 16%]
tests/test_transform_rules.py::test customer staging normalizes email and city and dedupes PASSED [ 33%]
tests/test_transform_rules.py::test order staging quarantines bad quantity and status PASSED [ 50%]
tests/test_transform_rules.py::test curated quarantines orphan customer without dropping silently PASSED [ 66%]
tests/test_transform_rules.py::test record hash is stable across pipeline runs PASSED [ 83%]
tests/test_transform_rules.py::test validate curated flags duplicate and out of range rows PASSED [100%]

===== 6 passed in 0.54s =====
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> git add tests/
>> git commit -m "test: add unit tests for transformation rules (verified passing)"
warning: in the working copy of 'tests/test_transform_rules.py', LF will be replaced by CRLF the next time Git touches it
[main fefaf9e] test: add unit tests for transformation rules (verified passing)
1 file changed, 126 insertions(+), 3 deletions(-)
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> python -m src.cli benchmark --repeats 5
storage_type file_size_bytes write_seconds full_read_seconds filtered_read_seconds row_count
csv 14947456 0.610800 0.160346 0.201958 49897
typed on read; no native schema/compression
json_lines 30639946 0.489718 0.483181 0.240503 49897
verbose (repeated keys); append/stream friendly
parquet 5456459 0.098763 0.051052 0.026300 49897
columnar + compressed; predicate/column pushdown on read
postgresql 16564224 NaN 0.287477 0.056027 49897 already loaded; write measured separately by
load timing; size is pg_total_relation_size (table+indexes+toast), not a single file
partitioned dataset written to C:\Users\Acer\Downloads\Project\dss150p-lab03-starter\data\partitioned
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> Get-Content data\benchmarks\benchmark_results.csv
storage_type,file_size_bytes,write_seconds,full_read_seconds,filtered_read_seconds,row_count,notes
csv,14947456,0.6107999000000746,0.16034639999998035,0.2019583999999668,49897,typed on read; no native schema/compression
json_lines,30639946,0.48971750000009706,0.4831807000000481,0.24050269999997909,49897,verbose (repeated keys); append/stream friendly
parquet,5456459,0.09876290000011068,0.05105200000025434,0.02629960000012943,49897,columnar + compressed; predicate/column pushdown on read
postgresql,16564224,0.2874772000000121,0.05602660000022297,49897,"already loaded; write measured separately by load timing; size is pg_total_r
elation_size (table+indexes+toast), not a single file"
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter>
```

The output shows that all 6 unit tests passed. The benchmark results are as follows:

storage_type	file_size_bytes	write_seconds	full_read_seconds	filtered_read_seconds	row_count	notes
csv	14947456	0.610800	0.160346	0.201958	49897	typed on read; no native schema/compression
json_lines	30639946	0.489718	0.483181	0.240503	49897	verbose (repeated keys); append/stream friendly
parquet	5456459	0.098763	0.051052	0.026300	49897	columnar + compressed; predicate/column pushdown on read
postgresql	16564224	NaN	0.287477	0.056027	49897	already loaded; write measured separately by load timing; size is pg_total_relation_size (table+indexes+toast), not a single file

Checkpoint 5 - load-partition run x2 (idempotent)

The screenshot shows a VS Code interface with a terminal window open. The terminal displays the output of several commands executed in a Python environment.

Test Results:

```
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> python -m pytest tests/ -v
rootdir: C:\Users\Acer\Downloads\Project\dss150p-lab03-starter
collected 6 items

tests/test_transform_rules.py::test dedupe latest keeps greatest updated at PASSED [ 16%]
tests/test_transform_rules.py::test customer staging normalizes email and city and dedupes PASSED [ 33%]
tests/test_transform_rules.py::test order staging quarantines bad quantity and status PASSED [ 50%]
tests/test_transform_rules.py::test curated quarantines orphan customer without dropping silently PASSED [ 66%]
tests/test_transform_rules.py::test record hash is stable across pipeline runs PASSED [ 83%]
tests/test_transform_rules.py::test validate curated flags duplicate and out of range rows PASSED [100%]

===== 6 passed in 0.54s =====
```

Git Commit:

```
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> git add tests/
>> git commit -m "test: add unit tests for transformation rules (verified passing)"
warning: in the working copy of 'tests/test_transform_rules.py', LF will be replaced by CRLF the next time Git touches it
[main fetafoe] test: add unit tests for transformation rules (verified passing)
1 file changed, 126 insertions(+), 3 deletions(-)
```

Benchmark Output:

```
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> python -m src.cli benchmark --repeats 5
storage_type  file_size_bytes  write_seconds  full_read_seconds  filtered_read_seconds  row_count
-----
csv           14947456          0.610880       0.160346           0.201958              49897
              notes
              typed on read; no native schema/compression
json_lines    30639946          0.489718       0.483181           0.240593              49897
              verbose (repeated keys); append/stream friendly
parquet       5456459           0.098763       0.051052           0.026300              49897
              columnar + compressed; predicate/column pushdown on read
postgresql    16564224         NaN            0.287477          0.056027              49897 already loaded; write measured separately by
load timing; size is pg total relation size (table+indexes+toast), not a single file
partitioned dataset written to c:\Users\Acer\Downloads\Project\dss150p-lab03-starter\data\partitioned
```

Get Content Command:

```
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> Get-Content data\benchmarks\benchmark_results.csv
storage_type,file_size_bytes,write_seconds,full_read_seconds,filtered_read_seconds,row_count,notes
csv,14947456,0.6107999000000746,0.16034639999998035,0.2019583999999668,49897,typed on read; no native schema/compression
json_lines,30639946,0.489717500000009706,0.4831807000000481,0.24050269999997909,49897,verbose (repeated keys); append/stream friendly
parquet,5456459,0.09876290000011068,0.05105200000025434,0.02629960000012943,49897,columnar + compressed; predicate/column pushdown on read
postgresql,16564224,0.2874772000000121,0.05602660000022297,0.056027,49897,"already loaded; write measured separately by load timing; size is pg total_r
elation size (table+indexes+toast), not a single file"
```

Copy-Item Command:

```
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> Copy-Item data\benchmarks\benchmark_results.csv docs\benchmark_results.csv -F
```

Load-Partition Command:

```
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> python -m src.cli load-partition --year 2026 --month 1
partition year=2026 month=1: rows_read=2506 rows_affected=0
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> python -m src.cli load-partition --year 2026 --month 1
partition year=2026 month=1: rows_read=2506 rows_affected=0
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter>
```

Checkpoint 5 - committed benchmark + partitioning code

The screenshot shows a VS Code editor with a terminal window open. The terminal displays the output of several commands executed in a Python environment. The commands and their outputs are as follows:

```
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> python -m pytest tests/ -v
tests/test_transform_rules.py::test_curated_quarantines_orphan_customer_without_dropping_silently PASSED
tests/test_transform_rules.py::test_record_hash_is_stable_across_pipeline_runs PASSED
tests/test_transform_rules.py::test_validate_curated_flags_duplicate_and_out_of_range_rows PASSED

===== 6 passed in 0.54s =====
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> git add tests/
>> git commit -m "test: add unit tests for transformation rules (verified passing)"
warning: in the working copy of 'tests/test_transform_rules.py', LF will be replaced by CRLF the next time Git touches it
[main fefaf9e] test: add unit tests for transformation rules (verified passing)
1 file changed, 126 insertions(+), 3 deletions(-)
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> python -m src.cli benchmark --repeats 5
storage_type file_size_bytes write_seconds full_read_seconds filtered_read_seconds row_count
-----
csv 14947456 0.610800 0.160346 0.201958 49897
typed on read; no native schema/compression
json_lines 30639946 0.489718 0.483181 0.240503 49897
verbose (repeated keys); append/stream friendly
parquet 5456459 0.098763 0.051852 0.026300 49897
columnar + compressed; predicate/column pushdown on read
postgresql 16564224 NaN 0.287477 0.056027 49897 already loaded; write measured separately by
load timing; size is pg_total_relation_size (tablesindexes+toast), not a single file
partitioned dataset written to C:\Users\Acer\Downloads\Project\dss150p-lab03-starter\data\partitioned
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> Get-Content data\benchmarks\benchmark_results.csv
storage_type,file_size_bytes,write_seconds,full_read_seconds,filtered_read_seconds,row_count,notes
csv,14947456,0.6107999999999999,0.16034639999999999,0.20195839999999999,49897,typed on read; no native schema/compression
json_lines,30639946,0.4897175000000000,0.4831807000000000,0.24050269999999999,49897,verbose (repeated keys); append/stream friendly
parquet,5456459,0.0987629000000000,0.0510520000000000,0.0262996000000000,49897,columnar + compressed; predicate/column pushdown on read
postgresql,16564224,0.2874772000000000,0.0560266000000000,0.0560272297,49897,"already loaded; write measured separately by load timing; size is pg_total_r
elation_size (tablesindexes+toast), not a single file"
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> Copy-Item data\benchmarks\benchmark_results.csv docs\benchmark_results.csv -F
once
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> python -m src.cli load-partition --year 2026 --month 1
partition year=2026 month=1: rows read=2506 rows affected=0
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> python -m src.cli load-partition --year 2026 --month 1
partition year=2026 month=1: rows read=2506 rows affected=0
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> git add src/benchmark docs/benchmark_results.csv
>> git commit -m "feat: storage benchmarking and Parquet partitioning (verified on real data)"
warning: in the working copy of 'src/benchmark/storage.py', LF will be replaced by CRLF the next time Git touches it
[main de317a2] feat: storage benchmarking and Parquet partitioning (verified on real data)
2 files changed, 161 insertions(+), 12 deletions(-)
create mode 100644 docs/benchmark_results.csv
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter>
```

The terminal output shows the results of a benchmark and partitioning test. The benchmark results are as follows:

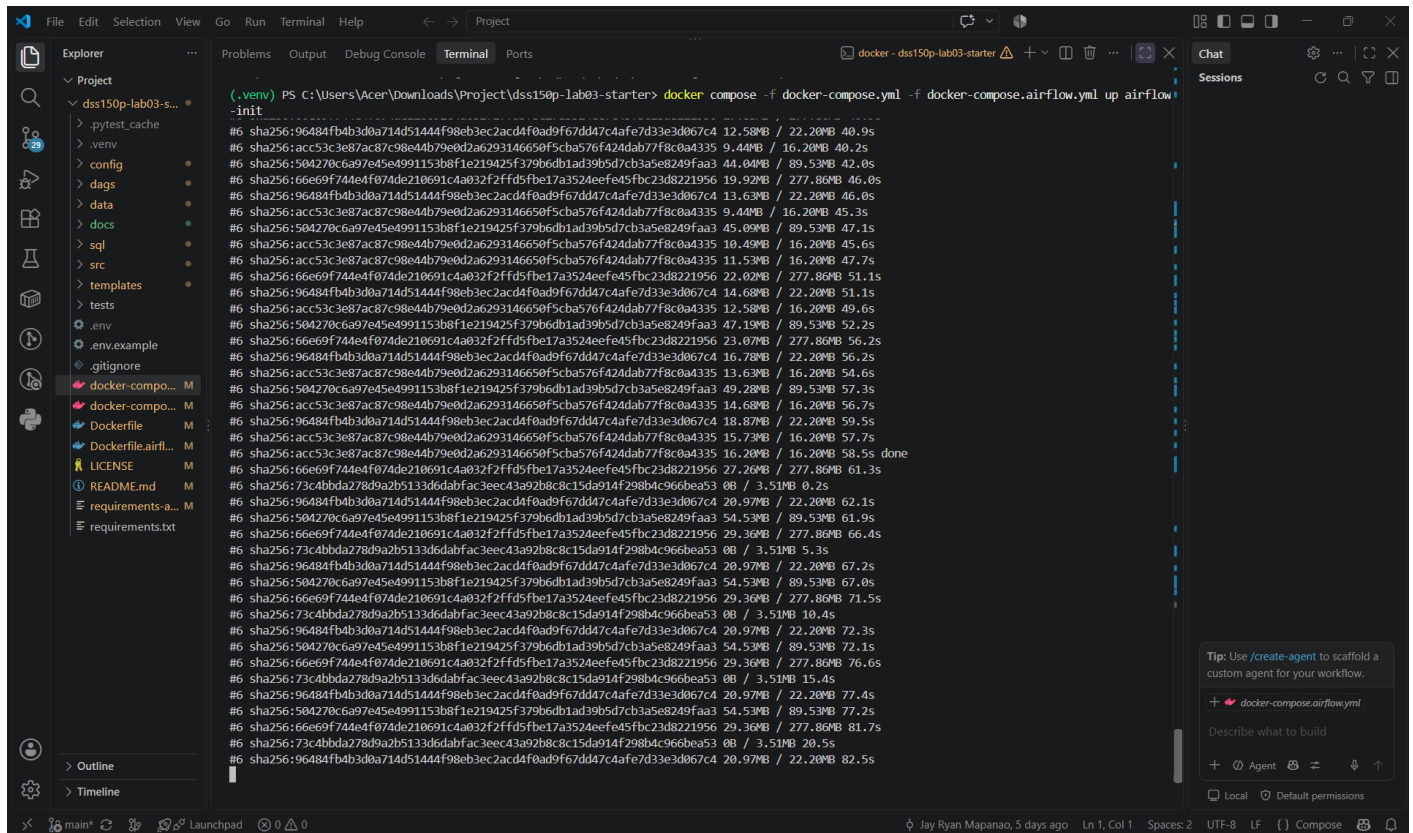
storage_type	file_size_bytes	write_seconds	full_read_seconds	filtered_read_seconds	row_count
csv	14947456	0.610800	0.160346	0.201958	49897
typed on read; no native schema/compression					
json_lines	30639946	0.489718	0.483181	0.240503	49897
verbose (repeated keys); append/stream friendly					
parquet	5456459	0.098763	0.051852	0.026300	49897
columnar + compressed; predicate/column pushdown on read					
postgresql	16564224	NaN	0.287477	0.056027	49897
already loaded; write measured separately by load timing; size is pg_total_relation_size (tablesindexes+toast), not a single file					

The partitioning test results are as follows:

```
partition year=2026 month=1: rows read=2506 rows affected=0
partition year=2026 month=1: rows read=2506 rows affected=0
```

The terminal also shows the execution of a benchmarking script and the creation of a new commit. The commit message is "feat: storage benchmarking and Parquet partitioning (verified on real data)".

Checkpoint 6 - Airflow image build (docker compose pulling images)

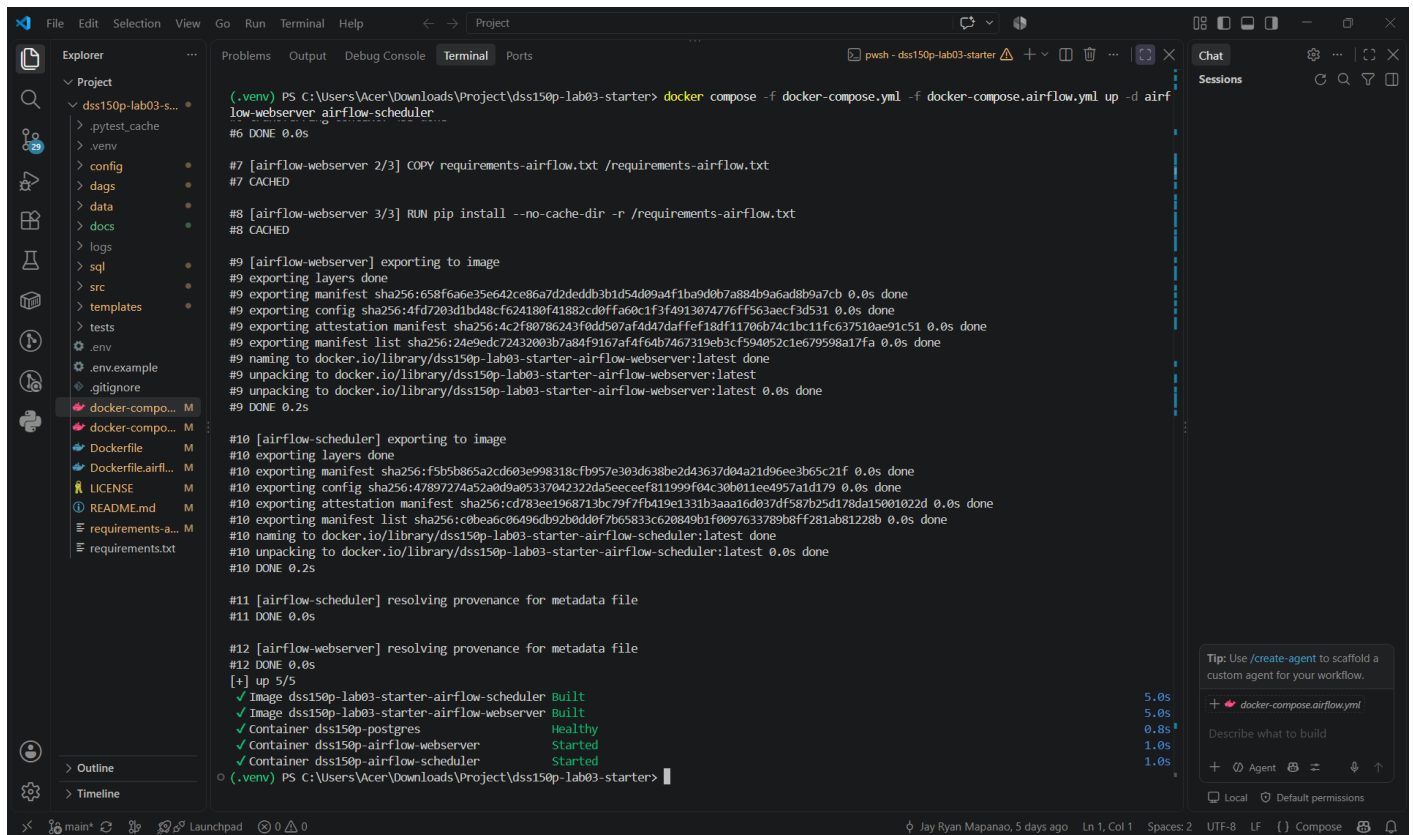


```
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> docker compose -f docker-compose.yml -f docker-compose.airflow.yml up airflow
-init
#6 sha256:96484fb4b3d8a714d51444f98eb3ec2acd4f0ad9f67dd47c4afe7d33e3d067c4 12.58MB / 22.20MB 40.9s
#6 sha256:acc53c3e87ac87c98e44b79e0d2a62931a6650f5cba576f424dab77f8c0a4335 0.44MB / 16.20MB 40.2s
#6 sha256:504270c6a97e45e4991153b8f1e219425f379b6db1ad39b5d7cb3a5e8249faa3 44.04MB / 89.53MB 42.0s
#6 sha256:66e69f744e4f074de210691c4a032f2ffdf5f17a3524eefe45fbc23d8221956 19.92MB / 277.86MB 46.0s
#6 sha256:96484fb4b3d8a714d51444f98eb3ec2acd4f0ad9f67dd47c4afe7d33e3d067c4 13.63MB / 22.20MB 46.0s
#6 sha256:acc53c3e87ac87c98e44b79e0d2a62931a6650f5cba576f424dab77f8c0a4335 0.44MB / 16.20MB 45.3s
#6 sha256:504270c6a97e45e4991153b8f1e219425f379b6db1ad39b5d7cb3a5e8249faa3 45.00MB / 89.53MB 47.1s
#6 sha256:acc53c3e87ac87c98e44b79e0d2a62931a6650f5cba576f424dab77f8c0a4335 10.49MB / 16.20MB 45.6s
#6 sha256:acc53c3e87ac87c98e44b79e0d2a62931a6650f5cba576f424dab77f8c0a4335 11.53MB / 16.20MB 47.7s
#6 sha256:66e69f744e4f074de210691c4a032f2ffdf5f17a3524eefe45fbc23d8221956 22.02MB / 277.86MB 51.1s
#6 sha256:96484fb4b3d8a714d51444f98eb3ec2acd4f0ad9f67dd47c4afe7d33e3d067c4 14.68MB / 22.20MB 51.1s
#6 sha256:acc53c3e87ac87c98e44b79e0d2a62931a6650f5cba576f424dab77f8c0a4335 12.58MB / 16.20MB 49.6s
#6 sha256:504270c6a97e45e4991153b8f1e219425f379b6db1ad39b5d7cb3a5e8249faa3 47.19MB / 89.53MB 52.2s
#6 sha256:66e69f744e4f074de210691c4a032f2ffdf5f17a3524eefe45fbc23d8221956 21.07MB / 277.86MB 56.2s
#6 sha256:96484fb4b3d8a714d51444f98eb3ec2acd4f0ad9f67dd47c4afe7d33e3d067c4 16.78MB / 22.20MB 56.2s
#6 sha256:acc53c3e87ac87c98e44b79e0d2a62931a6650f5cba576f424dab77f8c0a4335 13.63MB / 16.20MB 54.6s
#6 sha256:504270c6a97e45e4991153b8f1e219425f379b6db1ad39b5d7cb3a5e8249faa3 49.28MB / 89.53MB 57.3s
#6 sha256:acc53c3e87ac87c98e44b79e0d2a62931a6650f5cba576f424dab77f8c0a4335 14.68MB / 16.20MB 56.7s
#6 sha256:96484fb4b3d8a714d51444f98eb3ec2acd4f0ad9f67dd47c4afe7d33e3d067c4 18.78MB / 22.20MB 59.5s
#6 sha256:acc53c3e87ac87c98e44b79e0d2a62931a6650f5cba576f424dab77f8c0a4335 15.73MB / 16.20MB 57.7s
#6 sha256:504270c6a97e45e4991153b8f1e219425f379b6db1ad39b5d7cb3a5e8249faa3 54.53MB / 89.53MB 61.9s
#6 sha256:66e69f744e4f074de210691c4a032f2ffdf5f17a3524eefe45fbc23d8221956 29.36MB / 277.86MB 66.4s
#6 sha256:66e69f744e4f074de210691c4a032f2ffdf5f17a3524eefe45fbc23d8221956 27.26MB / 277.86MB 61.3s
#6 sha256:73c4bda278d9a2b5133d6dafac3eeca3a92b8c815da914f298b4c96bea53 08 / 3.51MB 0.2s
#6 sha256:96484fb4b3d8a714d51444f98eb3ec2acd4f0ad9f67dd47c4afe7d33e3d067c4 20.97MB / 22.20MB 62.1s
#6 sha256:504270c6a97e45e4991153b8f1e219425f379b6db1ad39b5d7cb3a5e8249faa3 54.53MB / 89.53MB 61.9s
#6 sha256:66e69f744e4f074de210691c4a032f2ffdf5f17a3524eefe45fbc23d8221956 29.36MB / 277.86MB 66.4s
#6 sha256:73c4bda278d9a2b5133d6dafac3eeca3a92b8c815da914f298b4c96bea53 08 / 3.51MB 5.3s
#6 sha256:96484fb4b3d8a714d51444f98eb3ec2acd4f0ad9f67dd47c4afe7d33e3d067c4 20.97MB / 22.20MB 67.2s
#6 sha256:504270c6a97e45e4991153b8f1e219425f379b6db1ad39b5d7cb3a5e8249faa3 54.53MB / 89.53MB 67.0s
#6 sha256:66e69f744e4f074de210691c4a032f2ffdf5f17a3524eefe45fbc23d8221956 29.36MB / 277.86MB 71.5s
#6 sha256:73c4bda278d9a2b5133d6dafac3eeca3a92b8c815da914f298b4c96bea53 08 / 3.51MB 10.4s
#6 sha256:96484fb4b3d8a714d51444f98eb3ec2acd4f0ad9f67dd47c4afe7d33e3d067c4 20.97MB / 22.20MB 72.3s
#6 sha256:504270c6a97e45e4991153b8f1e219425f379b6db1ad39b5d7cb3a5e8249faa3 54.53MB / 89.53MB 72.1s
#6 sha256:66e69f744e4f074de210691c4a032f2ffdf5f17a3524eefe45fbc23d8221956 29.36MB / 277.86MB 76.6s
#6 sha256:73c4bda278d9a2b5133d6dafac3eeca3a92b8c815da914f298b4c96bea53 08 / 3.51MB 15.4s
#6 sha256:96484fb4b3d8a714d51444f98eb3ec2acd4f0ad9f67dd47c4afe7d33e3d067c4 20.97MB / 22.20MB 77.4s
#6 sha256:504270c6a97e45e4991153b8f1e219425f379b6db1ad39b5d7cb3a5e8249faa3 54.53MB / 89.53MB 77.2s
#6 sha256:66e69f744e4f074de210691c4a032f2ffdf5f17a3524eefe45fbc23d8221956 29.36MB / 277.86MB 81.7s
#6 sha256:73c4bda278d9a2b5133d6dafac3eeca3a92b8c815da914f298b4c96bea53 08 / 3.51MB 20.5s
#6 sha256:96484fb4b3d8a714d51444f98eb3ec2acd4f0ad9f67dd47c4afe7d33e3d067c4 20.97MB / 22.20MB 82.5s
```

Checkpoint 6 - airflow-init completed, admin user created

```
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> docker compose -f docker-compose.yml -f docker-compose.airflow.yml up airflow-init
[2026-09-23T07:40:09.122+0000] [override.py:1912] INFO - Created Permission View: can edit on Users
[2026-09-23T07:40:09.131+0000] [override.py:1963] INFO - Added Permission can edit on Users to role Admin
[2026-09-23T07:40:09.140+0000] [override.py:1912] INFO - Created Permission View: can delete on Users
[2026-09-23T07:40:09.145+0000] [override.py:1963] INFO - Added Permission can delete on Users to role Admin
[2026-09-23T07:40:09.161+0000] [override.py:1912] INFO - Created Permission View: menu access on List Users
[2026-09-23T07:40:09.167+0000] [override.py:1963] INFO - Added Permission menu access on List Users to role Admin
[2026-09-23T07:40:09.182+0000] [override.py:1912] INFO - Created Permission View: menu access on Security
[2026-09-23T07:40:09.187+0000] [override.py:1963] INFO - Added Permission menu access on Security to role Admin
[2026-09-23T07:40:09.215+0000] [override.py:1912] INFO - Created Permission View: can create on Roles
[2026-09-23T07:40:09.221+0000] [override.py:1963] INFO - Added Permission can create on Roles to role Admin
[2026-09-23T07:40:09.228+0000] [override.py:1912] INFO - Created Permission View: can read on Roles
[2026-09-23T07:40:09.237+0000] [override.py:1963] INFO - Added Permission can read on Roles to role Admin
[2026-09-23T07:40:09.245+0000] [override.py:1912] INFO - Created Permission View: can edit on Roles
[2026-09-23T07:40:09.251+0000] [override.py:1963] INFO - Added Permission can edit on Roles to role Admin
[2026-09-23T07:40:09.260+0000] [override.py:1912] INFO - Created Permission View: can delete on Roles
[2026-09-23T07:40:09.267+0000] [override.py:1963] INFO - Added Permission can delete on Roles to role Admin
[2026-09-23T07:40:09.285+0000] [override.py:1912] INFO - Created Permission View: menu access on List Roles
[2026-09-23T07:40:09.291+0000] [override.py:1963] INFO - Added Permission menu access on List Roles to role Admin
[2026-09-23T07:40:09.315+0000] [override.py:1912] INFO - Created Permission View: can read on User Stats Chart
[2026-09-23T07:40:09.320+0000] [override.py:1963] INFO - Added Permission can read on User Stats Chart to role Admin
[2026-09-23T07:40:09.335+0000] [override.py:1912] INFO - Created Permission View: menu access on User's Statistics
[2026-09-23T07:40:09.341+0000] [override.py:1963] INFO - Added Permission menu access on User's Statistics to role Admin
[2026-09-23T07:40:09.369+0000] [override.py:1912] INFO - Created Permission View: can read on Permissions
[2026-09-23T07:40:09.374+0000] [override.py:1963] INFO - Added Permission can read on Permissions to role Admin
[2026-09-23T07:40:09.391+0000] [override.py:1912] INFO - Created Permission View: menu access on Actions
[2026-09-23T07:40:09.398+0000] [override.py:1963] INFO - Added Permission menu access on Actions to role Admin
[2026-09-23T07:40:09.426+0000] [override.py:1912] INFO - Created Permission View: can read on View Menus
[2026-09-23T07:40:09.431+0000] [override.py:1963] INFO - Added Permission can read on View Menus to role Admin
[2026-09-23T07:40:09.445+0000] [override.py:1912] INFO - Created Permission View: menu access on Resources
[2026-09-23T07:40:09.452+0000] [override.py:1963] INFO - Added Permission menu access on Resources to role Admin
[2026-09-23T07:40:09.482+0000] [override.py:1912] INFO - Created Permission View: can read on Permission Views
[2026-09-23T07:40:09.488+0000] [override.py:1963] INFO - Added Permission can read on Permission Views to role Admin
[2026-09-23T07:40:09.502+0000] [override.py:1912] INFO - Created Permission View: menu access on Permission Pairs
[2026-09-23T07:40:09.508+0000] [override.py:1963] INFO - Added Permission menu access on Permission Pairs to role Admin
[2026-09-23T07:40:10.413+0000] [override.py:1597] INFO - Added user admin
User "admin" created with role "Admin"
dss150p-airflow-init exited with code 0
```

Checkpoint 6 - airflow-webserver & airflow-scheduler containers started



```
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> docker compose -f docker-compose.yml -f docker-compose.airflow.yml up -d airflow-webserver airflow-scheduler
#6 DONE 0.0s

#7 [airflow-webserver 2/3] COPY requirements-airflow.txt /requirements-airflow.txt
#7 CACHED

#8 [airflow-webserver 3/3] RUN pip install --no-cache-dir -r /requirements-airflow.txt
#8 CACHED

#9 [airflow-webserver] exporting to image
#9 exporting layers done
#9 exporting manifest sha256:658f6a6e35e642ce86a7d2dedfb2b1d54d09a4f1ba9d0b7a884b9a6ad8b9a7cb 0.0s done
#9 exporting config sha256:4fd7203d1b4d8cf624180f41882cd0ffa60c1f3f4913074776ff563aecf3d531 0.0s done
#9 exporting attestation manifest sha256:4c2f80786243f0dd507af4d47daffaf18df11706b74c1bc11fc637510ae91c51 0.0s done
#9 exporting manifest list sha256:24e9edc72432003b7a84f9167af4f64b7467319eb3cf594052c1e679598a17fa 0.0s done
#9 naming to docker.io/library/dss150p-lab03-starter-airflow-webserver:latest done
#9 unpacking to docker.io/library/dss150p-lab03-starter-airflow-webserver:latest 0.0s done
#9 DONE 0.2s

#10 [airflow-scheduler] exporting to image
#10 exporting layers done
#10 exporting manifest sha256:f5b5b865a2cd603e998318cf1957e303d638be2d43637d04a21d96ee3b65c21f 0.0s done
#10 exporting config sha256:47897274a52a0d9a0533704222da5eeceef81199f04c30b01ee4957a1d179 0.0s done
#10 exporting attestation manifest sha256:cd783ee1968713bc79f7fb419e131b3aaa16d037df587b25d178da15001022d 0.0s done
#10 exporting manifest list sha256:c0bea6c06496db92b0dd0f7b65833c628849b1f0097633789b8ff281ab81228b 0.0s done
#10 naming to docker.io/library/dss150p-lab03-starter-airflow-scheduler:latest done
#10 unpacking to docker.io/library/dss150p-lab03-starter-airflow-scheduler:latest 0.0s done
#10 DONE 0.2s

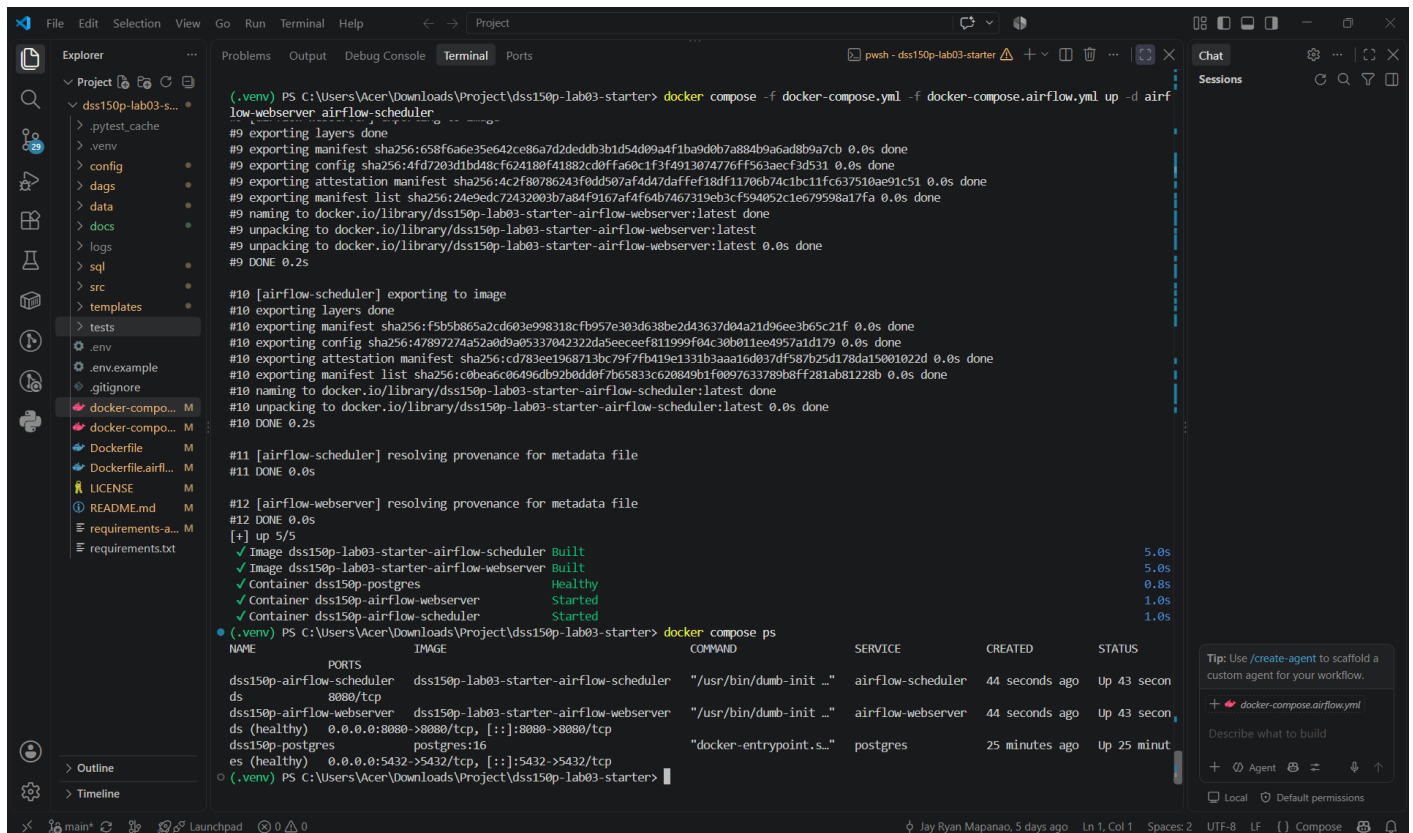
#11 [airflow-scheduler] resolving provenance for metadata file
#11 DONE 0.0s

#12 [airflow-webserver] resolving provenance for metadata file
#12 DONE 0.0s

[+] up 5/5
✓ Image dss150p-lab03-starter-airflow-scheduler Built 5.0s
✓ Image dss150p-lab03-starter-airflow-webserver Built 5.0s
✓ Container dss150p-postgres Healthy 0.8s
✓ Container dss150p-airflow-webserver Started 1.0s
✓ Container dss150p-airflow-scheduler Started 1.0s

(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter>
```

Checkpoint 6 - docker compose ps confirming all containers healthy



```
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> docker compose -f docker-compose.yml -f docker-compose.airflow.yml up -d airflow-webserver airflow-scheduler
#9 exporting layers done
#9 exporting manifest sha256:658f6a6e35e642ce86a7d2dedb3b1d54d09a4f1ba9d0b7a884b9a6ad8b9a7cb 0.0s done
#9 exporting config sha256:4fd7203d1bd48cf62a180f41882cd0ffa60c1f3f4913074776ff563aecf3d511 0.0s done
#9 exporting attestation manifest sha256:4c2f80786243f0dd507af4d47daffef18df11706b74c1bc11fc637510ae91c51 0.0s done
#9 exporting manifest list sha256:24e9edc72432003b7a84f9167af4f64b7467319eb3cf94052c1e679598a17fa 0.0s done
#9 naming to docker.io/library/dss150p-lab03-starter-airflow-webserver:latest done
#9 unpacking to docker.io/library/dss150p-lab03-starter-airflow-webserver:latest done
#9 unpacking to docker.io/library/dss150p-lab03-starter-airflow-webserver:latest 0.0s done
#9 DONE 0.2s

#10 [airflow-scheduler] exporting to image
#10 exporting layers done
#10 exporting manifest sha256:f5b5b865a2cd603e998318cf1957e303d638be2d43637d04a21d96ee3b65c21f 0.0s done
#10 exporting config sha256:47897274a52a0d9a05337042322da5eeceef81199f04c30b011ee4957a1d179 0.0s done
#10 exporting attestation manifest sha256:cd783ee1068713bc79f7fb419e1331b3aaa16d037df587b25d178da15001022d 0.0s done
#10 exporting manifest list sha256:c0ba6c0649c6b92b0d0f7b65833c620809b1f097633789b8ff281ab81228b 0.0s done
#10 naming to docker.io/library/dss150p-lab03-starter-airflow-scheduler:latest done
#10 unpacking to docker.io/library/dss150p-lab03-starter-airflow-scheduler:latest 0.0s done
#10 DONE 0.2s

#11 [airflow-scheduler] resolving provenance for metadata file
#11 DONE 0.0s

#12 [airflow-webserver] resolving provenance for metadata file
#12 DONE 0.0s

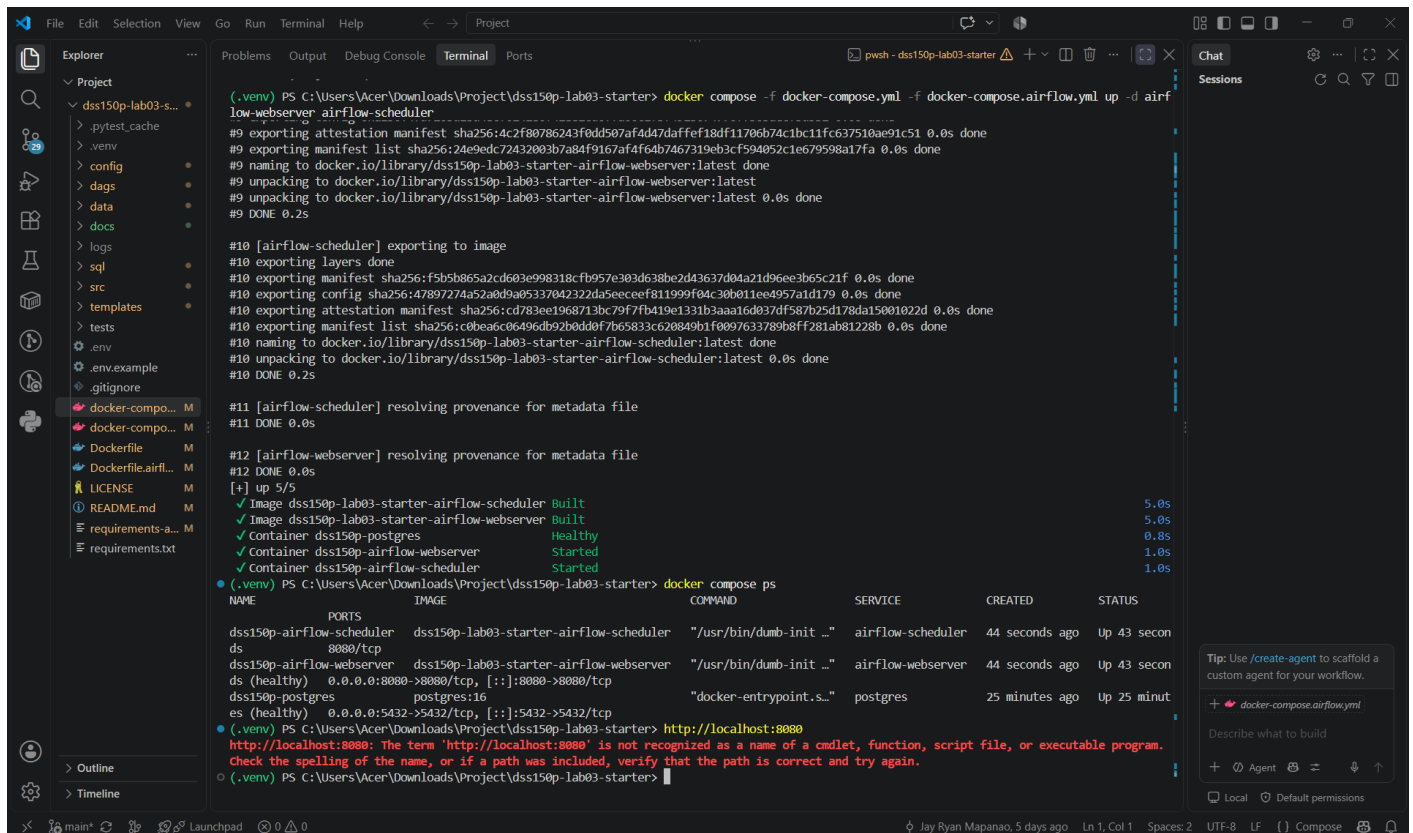
[+] up 5/5
✓ Image dss150p-lab03-starter-airflow-scheduler Built 5.0s
✓ Image dss150p-lab03-starter-airflow-webserver Built 5.0s
✓ Container dss150p-postgres Healthy 0.8s
✓ Container dss150p-airflow-webserver Started 1.0s
✓ Container dss150p-airflow-scheduler Started 1.0s

(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> docker compose ps
```

NAME	PORTS	IMAGE	COMMAND	SERVICE	CREATED	STATUS
dss150p-airflow-scheduler	8080/tcp	dss150p-lab03-starter-airflow-scheduler	"/usr/bin/dumb-init ..."	airflow-scheduler	44 seconds ago	Up 43 secon
dss150p-airflow-webserver	0.0.0.0:8080->8080/tcp, [::]:8080->8080/tcp	dss150p-lab03-starter-airflow-webserver	"/usr/bin/dumb-init ..."	airflow-webserver	44 seconds ago	Up 43 secon
dss150p-postgres	postgres:16	postgres:16	"docker-entrypoint.s..."	postgres	25 minutes ago	Up 25 minut

```
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter>
```

Checkpoint 6 - confirming containers up (localhost:8080 typed in terminal by mistake)



```
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> docker compose -f docker-compose.yml -f docker-compose.airflow.yml up -d airflow
low-webserver airflow-scheduler
#9 exporting attestation manifest sha256:4c2f80786243f0dd507af4d47daffef18df11706b74c1bc11fc637510ae91c51 0.0s done
#9 exporting manifest list sha256:24e9edc72432003b7a84f9167af4f64b7467319eb3cf594052c1e679598a17fa 0.0s done
#9 naming to docker.io/library/dss150p-lab03-starter-airflow-webserver:latest done
#9 unpacking to docker.io/library/dss150p-lab03-starter-airflow-webserver:latest 0.0s done
#9 unpacking to docker.io/library/dss150p-lab03-starter-airflow-webserver:latest 0.0s done
#9 DONE 0.2s

#10 [airflow-scheduler] exporting to image
#10 exporting layers done
#10 exporting manifest sha256:f5b5b865a2cd603e998318cfb957e303d638be2d43637d04a21d96ee3b65c21f 0.0s done
#10 exporting config sha256:47897774a52ad0a05337042322da5eeceef811999f04c30b011ee4957a1d179 0.0s done
#10 exporting attestation manifest sha256:cd783ee1068713bc79f7fb419e1331b3aaa16d037df587b25d178da15001022d 0.0s done
#10 exporting manifest list sha256:c0ba6c0649c6b92b0d40f7b65833c620849b1f0097633789b8ff281ab81228b 0.0s done
#10 naming to docker.io/library/dss150p-lab03-starter-airflow-scheduler:latest done
#10 unpacking to docker.io/library/dss150p-lab03-starter-airflow-scheduler:latest 0.0s done
#10 DONE 0.2s

#11 [airflow-scheduler] resolving provenance for metadata file
#11 DONE 0.0s

#12 [airflow-webserver] resolving provenance for metadata file
#12 DONE 0.0s
[+] up 5/5
✔ Image dss150p-lab03-starter-airflow-scheduler Built 5.0s
✔ Image dss150p-lab03-starter-airflow-webserver Built 5.0s
✔ Container dss150p-postgres Healthy 0.8s
✔ Container dss150p-airflow-webserver Started 1.0s
✔ Container dss150p-airflow-scheduler Started 1.0s

(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> docker compose ps

```

NAME	IMAGE	COMMAND	SERVICE	CREATED	STATUS
dss150p-airflow-scheduler	dss150p-lab03-starter-airflow-scheduler	"/usr/bin/dumb-init _"	airflow-scheduler	44 seconds ago	Up 43 secon
ds	8080/tcp				
dss150p-airflow-webserver	dss150p-lab03-starter-airflow-webserver	"/usr/bin/dumb-init _"	airflow-webserver	44 seconds ago	Up 43 secon
ds (healthy)	0.0.0.0:8080->8080/tcp, [::]:8080->8080/tcp				
dss150p-postgres	postgres:16	"docker-entrypoint.s"	postgres	25 minutes ago	Up 25 minut
es (healthy)	0.0.0.0:5432->5432/tcp, [::]:5432->5432/tcp				

```
(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter> http://localhost:8080
http://localhost:8080: The term 'http://localhost:8080' is not recognized as a name of a cmdlet, function, script file, or executable program.
Check the spelling of the name, or if a path was included, verify that the path is correct and try again.

(.venv) PS C:\Users\Acer\Downloads\Project\dss150p-lab03-starter>
```

Checkpoint 6 - Airflow UI: DAG dss150p_sales_pipeline listed, no import errors

 Airflow

DAGs

Cluster Activity

Datasets

Security

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Admin

Docs

 07:42 UTC 

DAGs

All 1Active 0Paused 1

Running 0Failed 0

Filter DAGs by tag

Search DAGs

☒ Auto-refresh 

DAG	Owner	Runs	Schedule	Last Run	Next Run	Recent Tasks	Actions	Links
☐ dss150p_sales_pipeline DSS150P	dss150p		02*** 		2026-09-22, 02:00:00 		 ...	

«

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Showing 1-1 of 1 DAGs

Version: v2.10.5

Git Version: .release:b93c3db6b1641b0840bd15ac7d05bc58ff2cccbf

Checkpoint 6 - Trigger DAG w/ config form (run_mode=full)

 Airflow

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 07:44 UTC 

Trigger DAG: dss150p_sales_pipeline

DAG conf Parameters

run_mode:

full

×

▼

year *:

2026

month *:

1

Generated Configuration JSON and Dagrun Options

Trigger

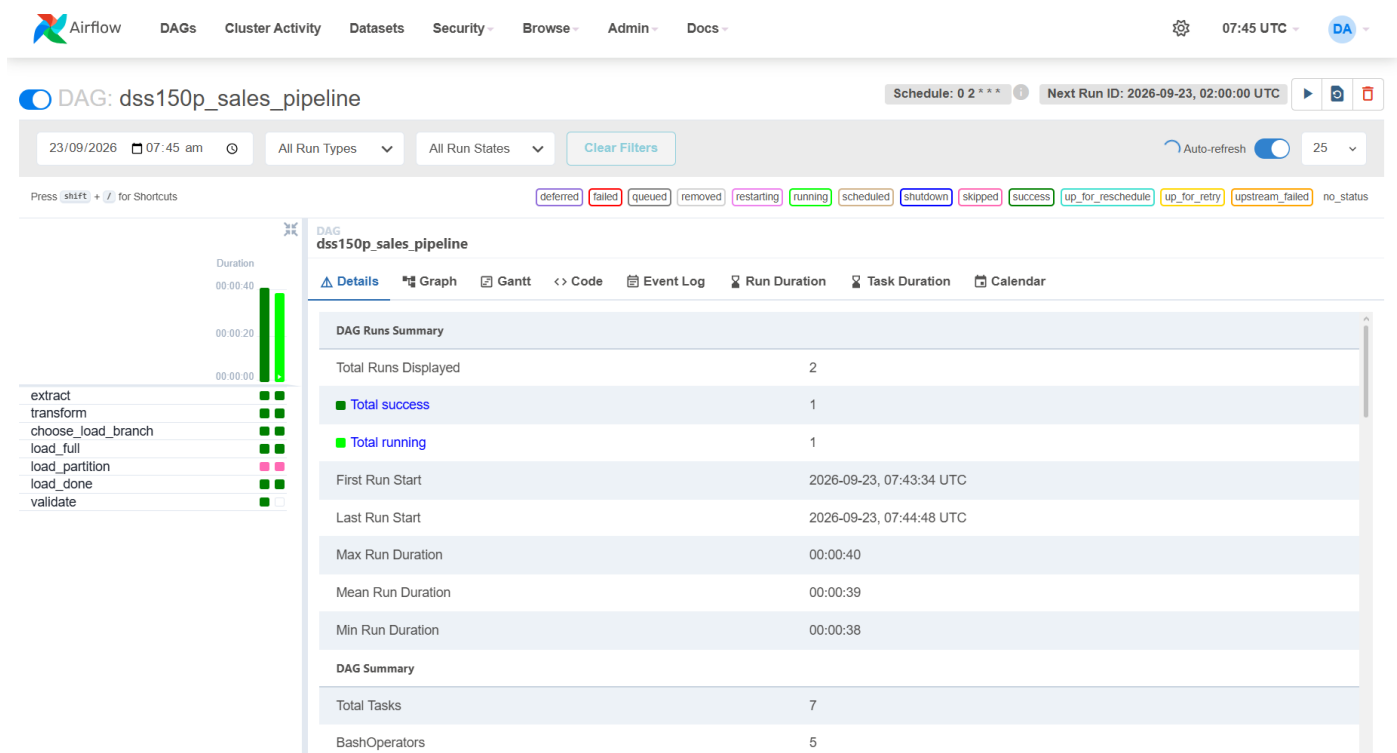
Cancel

Version: v2.10.5

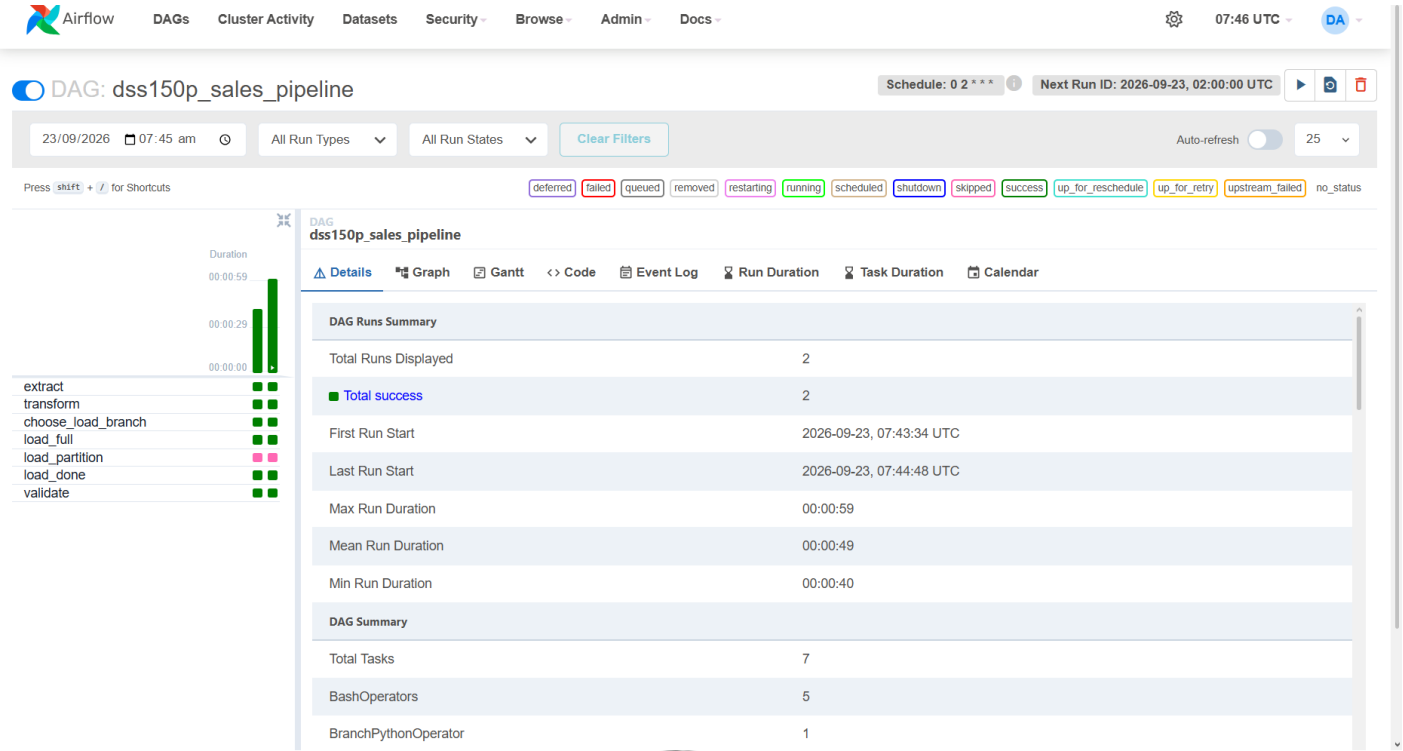
Git Version: .release:b93c3db6b1641b0840bd15ac7d05bc58ff2cccbf

DSS150P Lab 3

Checkpoint 6 - Grid view: full-mode run in progress



Checkpoint 6 - Grid view: 2 full-mode runs succeeded (load_partition correctly skipped)



Checkpoint 6 - Trigger DAG w/ config (run_mode=partition, year=2026, month=2)

 Airflow

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 07:47 UTC 

month *

1

Generated Configuration JSON and Dagrun Options

Logical date:

 2026-09-23T07:46:57+00:00

Run id:

Run id, optional - will be generated if not provided

```
1. {"run_mode": "partition", "year": 2026, "month": 2}
```

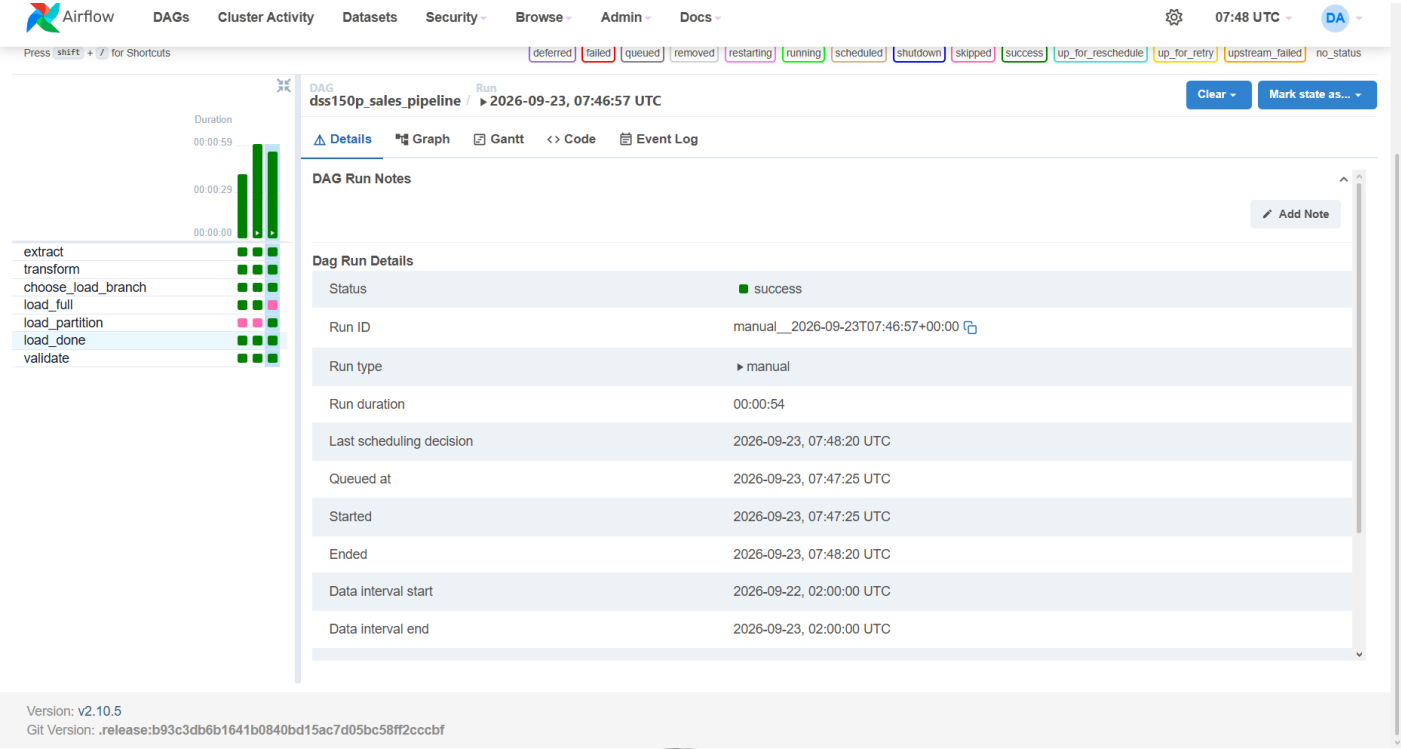
Trigger

Cancel

Version: v2.10.5

Git Version: .release:b93c3db6b1641b0840bd15ac7d05bc58ff2cccbf

Checkpoint 6 - Grid view: partition-mode run succeeded (load_full correctly skipped)

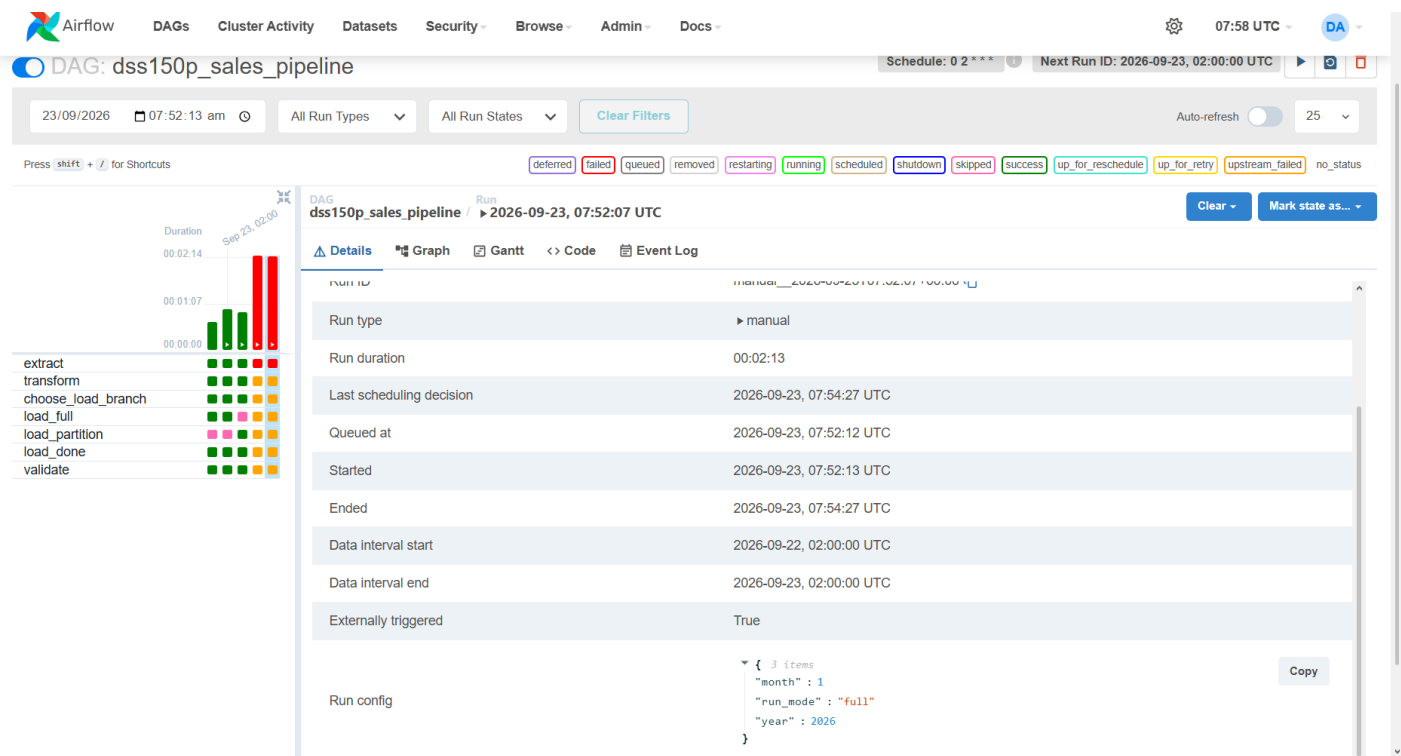


Checkpoint 7 - CSRF token error (session timeout, retried trigger)

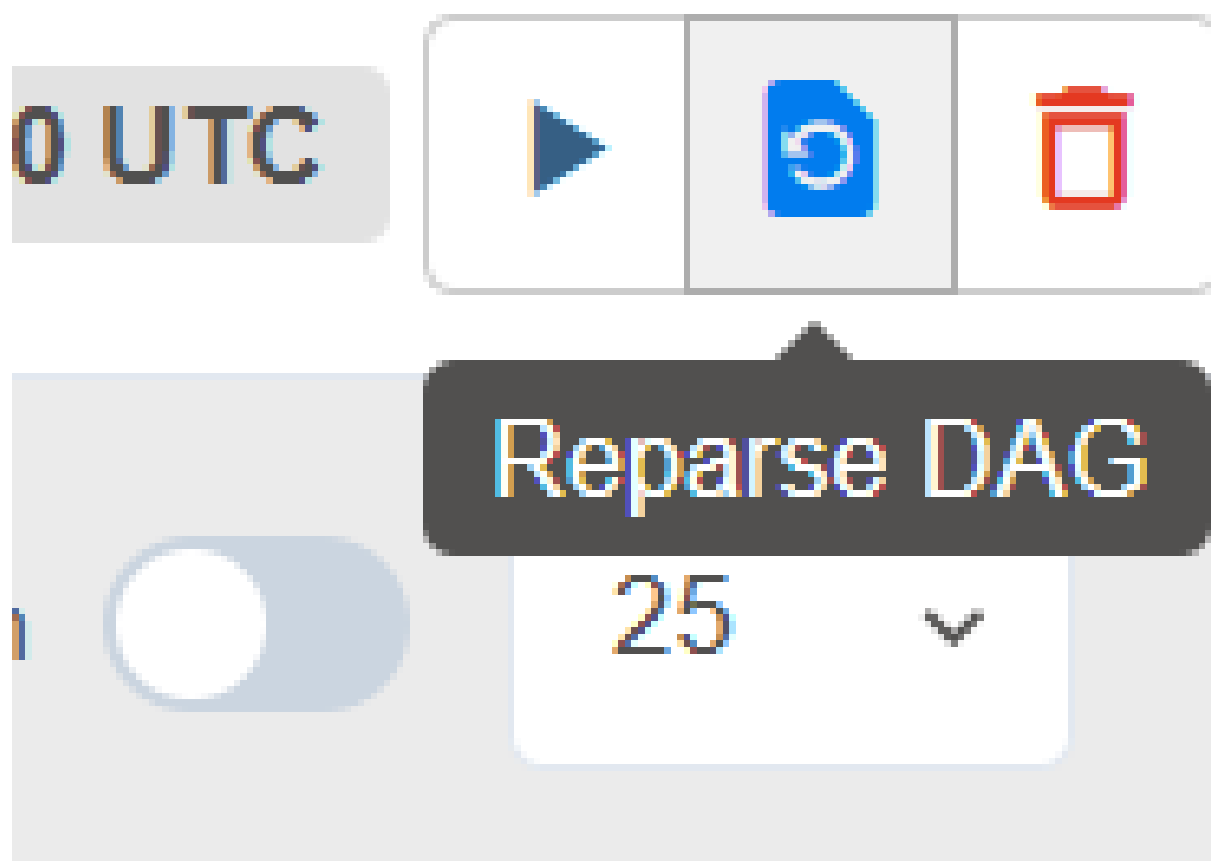
Bad Request

The CSRF token is missing.

Checkpoint 7 - Deliberate failure: extract task failed after retries (upstream_failed cascade)

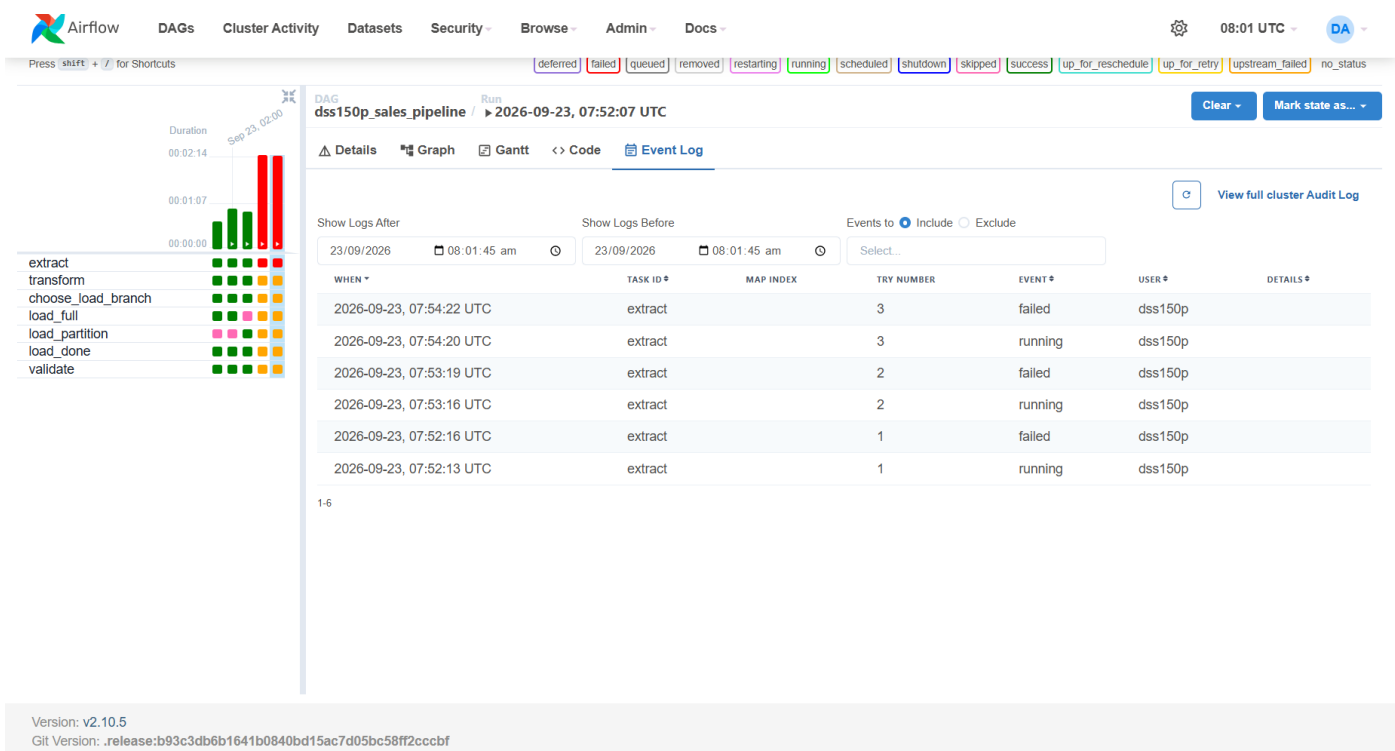


Checkpoint 7 - Locating task log via Grid view (UI navigation)



stream failed | no status

Checkpoint 7 - Event Log showing 3 retry attempts (try 1, 2, 3) matching retries=2 config



Checkpoint 7 - Recovery run succeeded after restoring orders.csv (no manual DB cleanup needed)

